

# Gastrointestinal

*“A good set of bowels is worth more to a man than any quantity of brains.”*  
—Josh Billings

*“Man should strive to have his intestines relaxed all the days of his life.”*  
—Moses Maimonides

*“Is life worth living? It all depends on the liver.”*  
—William James

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## ► GASTROINTESTINAL—EMBRYOLOGY

**Normal gastrointestinal embryology**

Foregut—esophagus to upper duodenum.

Midgut—lower duodenum to proximal  $\frac{2}{3}$  of transverse colon.

Hindgut—distal  $\frac{1}{3}$  of transverse colon to anal canal above pectinate line.

Midgut development:

- 6th week—physiologic midgut herniates through umbilical ring
- 10th week—returns to abdominal cavity + rotates around superior mesenteric artery (SMA), total  $270^\circ$  counterclockwise

**Ventral wall defects and hernias**

Developmental defects due to failure of:

- Rostral fold closure—sternal defects (ectopia cordis)
- Lateral fold closure—omphalocele, gastroschisis
- Caudal fold closure—bladder exstrophy

**Gastroschisis**—extrusion of abdominal contents through abdominal folds (typically right of umbilicus); not covered by peritoneum or amnion.

**Omphalocele**—persistent herniation of abdominal contents into umbilical cord, **sealed** by peritoneum **A**.

**Congenital umbilical hernia**—incomplete closure of umbilical ring. Many close spontaneously.

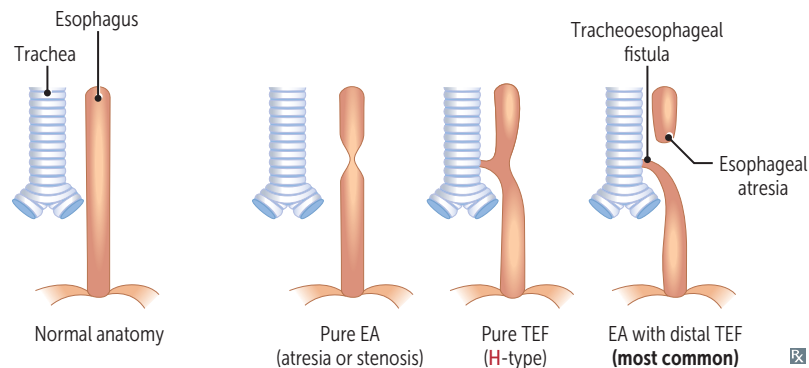
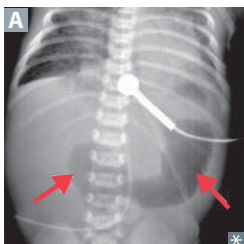
**Tracheoesophageal anomalies**

Esophageal atresia (EA) with distal tracheoesophageal fistula (TEF) is the most common (85%).

Polyhydramnios in utero. Neonates drool, choke, and vomit with first feeding. TEF allows air to enter stomach (visible on CXR). Cyanosis is  $2^\circ$  to laryngospasm (to avoid reflux-related aspiration).

Clinical test: failure to pass nasogastric tube into stomach.

In **H-type**, the fistula resembles the letter **H**. In pure EA the CXR shows gasless abdomen.

**Intestinal atresia**

Presents with bilious vomiting and abdominal distension within first 1–2 days of life.

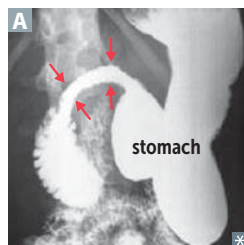
**Duodenal atresia**—failure to recanalize. Associated with “double bubble” (dilated stomach, proximal duodenum) on x-ray **A**). Associated with Down syndrome.

**Jejunal and ileal atresia**—disruption of mesenteric vessels → ischemic necrosis → segmental resorption (bowel discontinuity or “apple peel”).

### Hypertrophic pyloric stenosis

Most common cause of gastric outlet obstruction in infants (1:600). Palpable olive-shaped mass in epigastric region, visible peristaltic waves, and nonbilious projectile vomiting at ~ 2–6 weeks old. More common in firstborn males; associated with exposure to macrolides. Results in hypokalemic hypochloremic metabolic alkalosis (2° to vomiting of gastric acid and subsequent volume contraction). Treatment is surgical incision (pyloromyotomy).

### Pancreas and spleen embryology

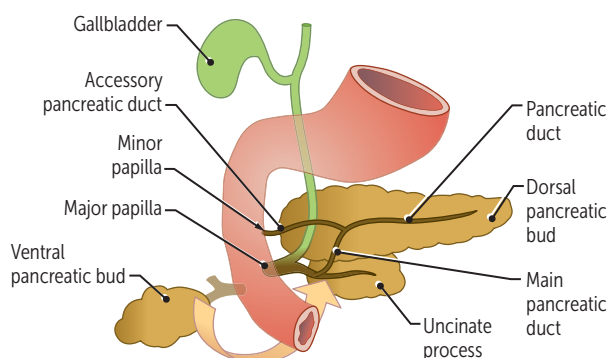


Pancreas—derived from foregut. Ventral pancreatic buds contribute to uncinate process and main pancreatic duct. The dorsal pancreatic bud alone becomes the body, tail, isthmus, and accessory pancreatic duct. Both the ventral and dorsal buds contribute to pancreatic head.

**Annular pancreas**—ventral pancreatic bud abnormally encircles 2nd part of duodenum; forms a ring of pancreatic tissue that may cause duodenal narrowing **A** and vomiting.

**Pancreas divisum**—ventral and dorsal parts fail to fuse at 8 weeks. Common anomaly; mostly asymptomatic, but may cause chronic abdominal pain and/or pancreatitis.

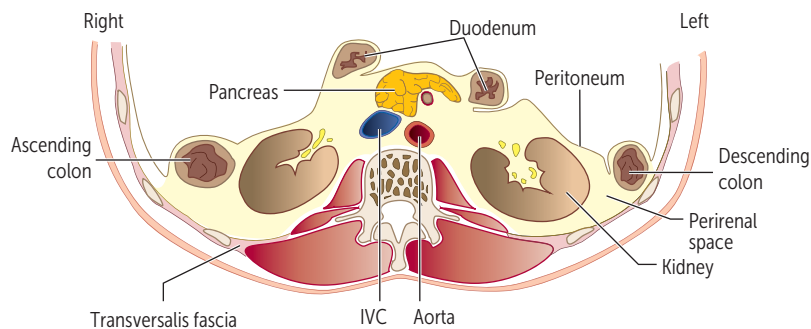
Spleen—arises in mesentery of stomach (hence is mesodermal) but has foregut supply (celiac trunk → splenic artery).



## ► GASTROINTESTINAL—ANATOMY

### Retroperitoneal structures

Retroperitoneal structures include GI structures that lack a mesentery and non-GI structures. Injuries to retroperitoneal structures can cause blood or gas accumulation in retroperitoneal space.



### SAD PUCKER:

Suprarenal (adrenal) glands [not shown]

Aorta and IVC

Duodenum (2nd through 4th parts)

Pancreas (except tail)

Ureters [not shown]

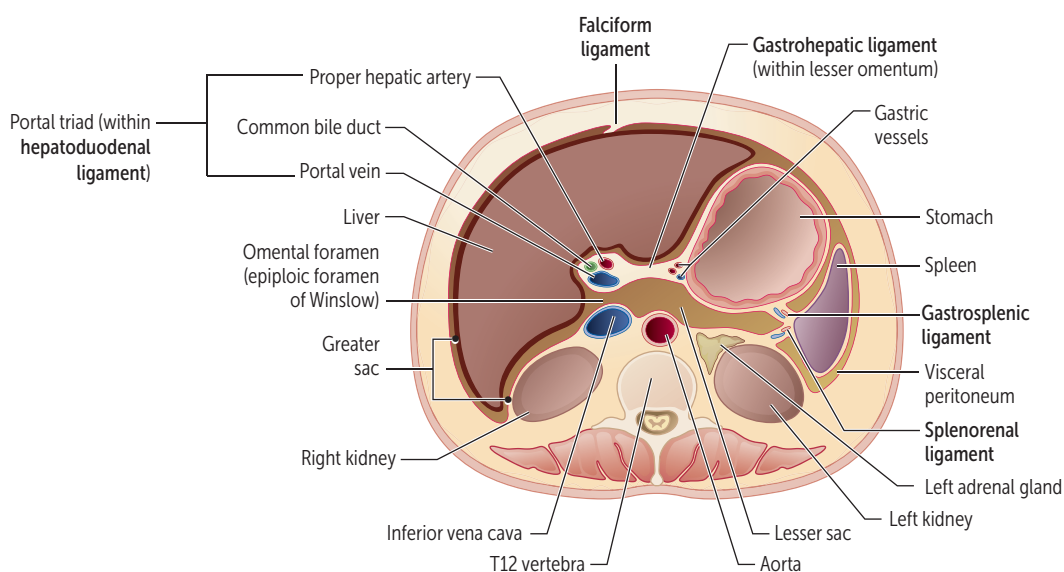
Colon (descending and ascending)

Kidneys

Esophagus (thoracic portion) [not shown]

Rectum (partially) [not shown]

## Important gastrointestinal ligaments



| LIGAMENT                       | CONNECTS                               | STRUCTURES CONTAINED                                               | NOTES                                                                                                                                                                                                                         |
|--------------------------------|----------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Falciform</b>               | Liver to anterior abdominal wall       | Ligamentum teres hepatis (derivative of fetal umbilical vein)      | Derivative of ventral mesentery                                                                                                                                                                                               |
| <b>Hepatoduodenal</b>          | Liver to duodenum                      | Portal triad: proper hepatic artery, portal vein, common bile duct | Pringle maneuver—ligament may be compressed between thumb and index finger placed in omental foramen to control bleeding<br>Borders the omental foramen, which connects the greater and lesser sacs<br>Part of lesser omentum |
| <b>Gastrohepatic</b>           | Liver to lesser curvature of stomach   | Gastric arteries                                                   | Separates greater and lesser sacs on the right<br>May be cut during surgery to access lesser sac<br>Part of lesser omentum                                                                                                    |
| <b>Gastrocolic</b> (not shown) | Greater curvature and transverse colon | Gastroepiploic arteries                                            | Part of greater omentum                                                                                                                                                                                                       |
| <b>Gastrosplenic</b>           | Greater curvature and spleen           | Short gastrics, left gastroepiploic vessels                        | Separates greater and lesser sacs on the left<br>Part of greater omentum                                                                                                                                                      |
| <b>Splenorenal</b>             | Spleen to posterior abdominal wall     | Splenic artery and vein, tail of pancreas                          |                                                                                                                                                                                                                               |

## Digestive tract anatomy

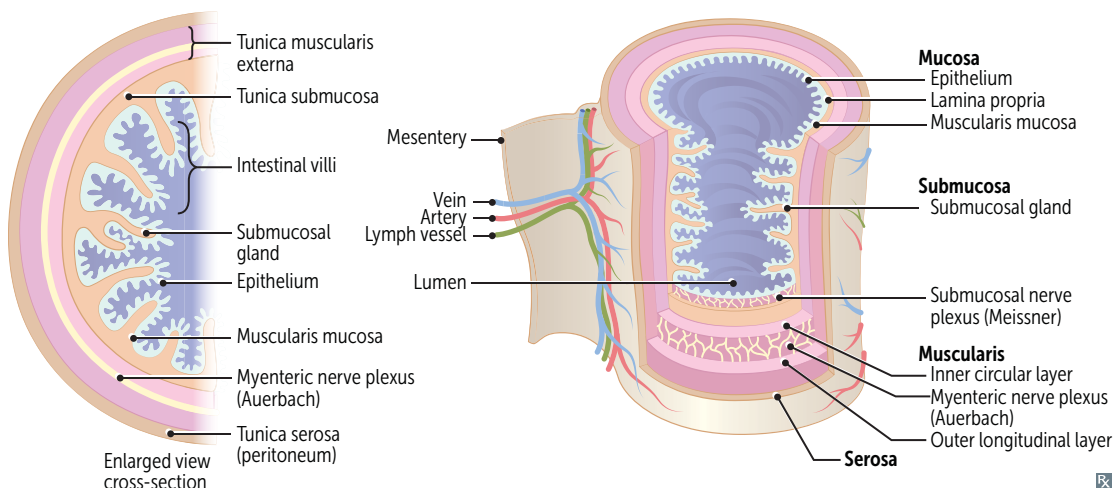
Layers of gut wall (inside to outside—**MSMS**):

- **M**ucosa—epithelium, lamina propria, muscularis mucosa
- **S**ubmucosa—includes **S**ubmucosal nerve plexus (Meissner), **S**ecretes fluid
- **M**uscularis externa—includes **M**yenteric nerve plexus (Auerbach), **M**otility
- **S**erosa (when intraperitoneal), adventitia (when retroperitoneal)

Ulcers can extend into submucosa, inner or outer muscular layer. Erosions are in the mucosa only.

Frequencies of basal electric rhythm (slow waves):

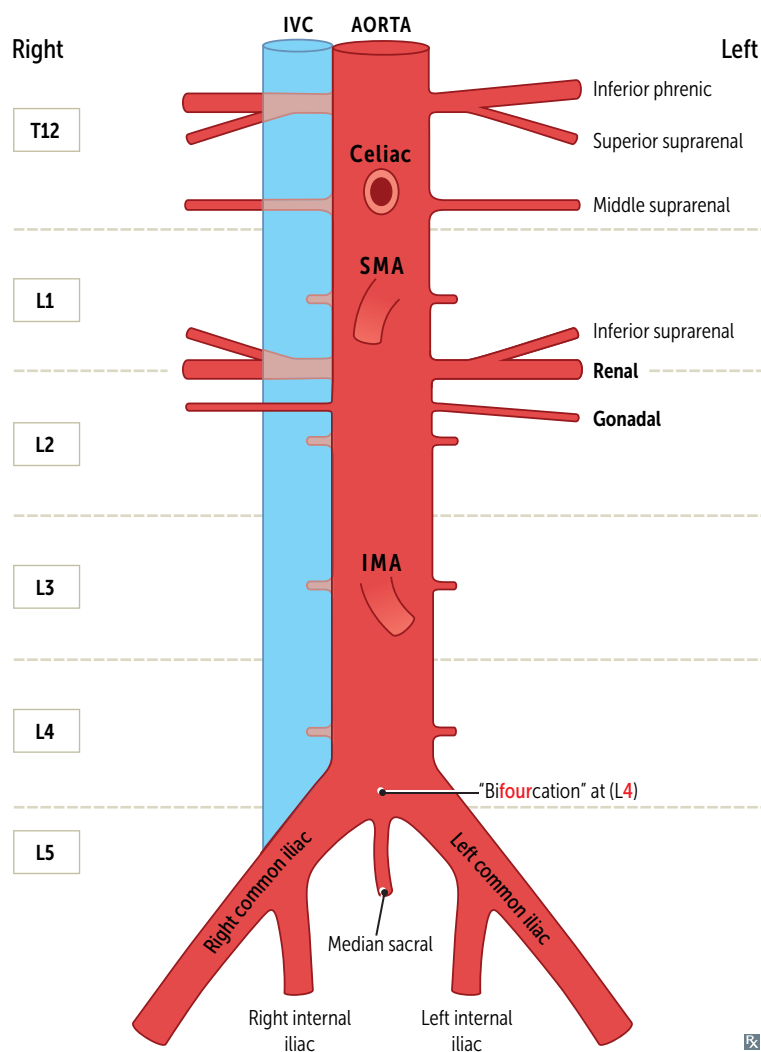
- Stomach—3 waves/min
- Duodenum—12 waves/min
- Ileum—8–9 waves/min



## Digestive tract histology

|                  |                                                                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Esophagus</b> | Nonkeratinized stratified squamous epithelium.                                                                                                                                                                                                                 |
| <b>Stomach</b>   | Gastric glands.                                                                                                                                                                                                                                                |
| <b>Duodenum</b>  | Villi and microvilli ↑ absorptive surface.<br>Brunner glands ( $\text{HCO}_3^-$ -secreting cells of submucosa) and crypts of Lieberkühn (contain stem cells that replace enterocytes/goblet cells and Paneth cells that secrete defensins, lysozyme, and TNF). |
| <b>Jejunum</b>   | Plicae circulares (also present in distal duodenum) and crypts of Lieberkühn.                                                                                                                                                                                  |
| <b>Ileum</b>     | Peyer patches (lymphoid aggregates in lamina propria, submucosa), plicae circulares (proximal ileum), and crypts of Lieberkühn.<br>Largest number of goblet cells in the small intestine.                                                                      |
| <b>Colon</b>     | Crypts of Lieberkühn but no villi; abundant goblet cells.                                                                                                                                                                                                      |

### Abdominal aorta and branches



Arteries supplying GI structures branch **anteriorly**. Arteries supplying non-GI structures branch **laterally** and **posteriorly**.

#### Superior mesenteric artery syndrome—

Characterized by intermittent intestinal obstruction symptoms (primarily postprandial pain) when transverse (third) portion of duodenum is compressed between SMA and aorta. Typically occurs in conditions associated with diminished mesenteric fat (eg, low body weight/malnutrition).

### Gastrointestinal blood supply and innervation

| EMBRYONIC GUT REGION | ARTERY | PARASYMPATHETIC INNERVATION | VERTEBRAL LEVEL | STRUCTURES SUPPLIED                                                                                                                       |
|----------------------|--------|-----------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Foregut</b>       | Celiac | Vagus                       | T12/L1          | Pharynx (vagus nerve only) and lower esophagus (celiac artery only) to proximal duodenum; liver, gallbladder, pancreas, spleen (mesoderm) |
| <b>Midgut</b>        | SMA    | Vagus                       | L1              | Distal duodenum to proximal $\frac{2}{3}$ of transverse colon                                                                             |
| <b>Hindgut</b>       | IMA    | Pelvic                      | L3              | Distal $\frac{1}{3}$ of transverse colon to upper portion of rectum                                                                       |

**Celiac trunk**

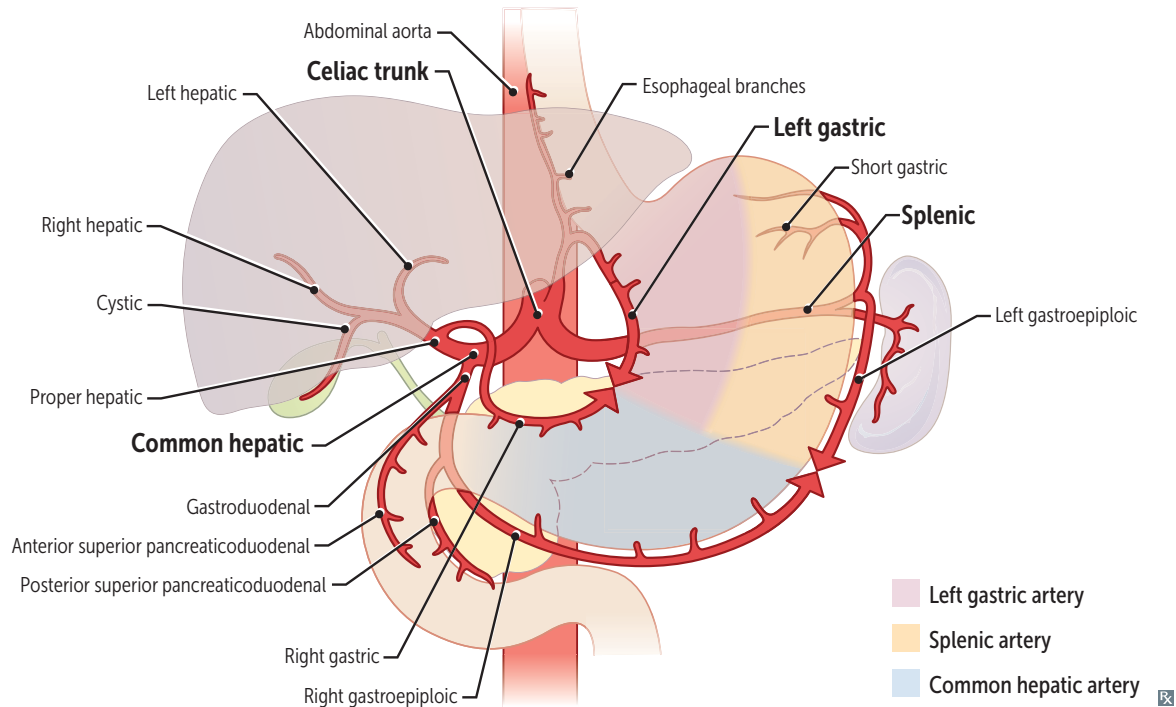
Branches of celiac trunk: common hepatic, splenic, and left gastric. These constitute the main blood supply of the stomach.

Strong anastomoses exist between:

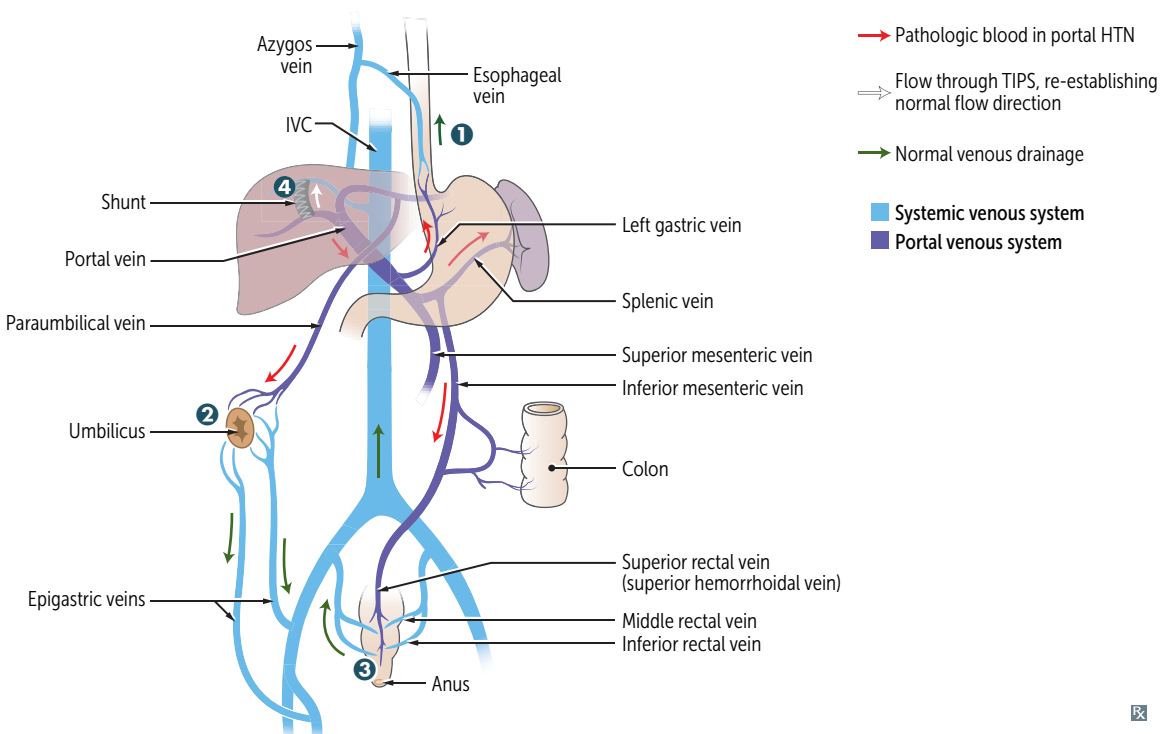
- Left and right gastroepiploics
- Left and right gastrics

Posterior duodenal ulcers penetrate gastroduodenal artery causing hemorrhage.

Anterior duodenal ulcers perforate into the anterior abdominal cavity, potentially leading to pneumoperitoneum.



### Portosystemic anastomoses



| SITE OF ANASTOMOSIS | CLINICAL SIGN        | PORTAL ↔ SYSTEMIC                                                      |
|---------------------|----------------------|------------------------------------------------------------------------|
| ① <b>Esophagus</b>  | Esophageal varices   | Left gastric ↔ azygos                                                  |
| ② <b>Umbilicus</b>  | <b>Caput</b> medusae | Paraumbilical ↔ small epigastric veins of the anterior abdominal wall. |
| ③ <b>Rectum</b>     | Anorectal varices    | Superior rectal ↔ middle and inferior rectal                           |

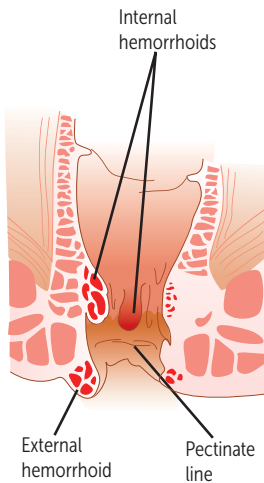
Varices of **gut**, **butt**, and **caput** (medusae) are commonly seen with portal hypertension.

- ④ Treatment with a **transjugular intrahepatic portosystemic shunt (TIPS)** between the portal vein and hepatic vein relieves portal hypertension by shunting blood to the systemic circulation, bypassing the liver. Can precipitate hepatic encephalopathy.



### Pectinate (dentate) line

Formed where endoderm (hindgut) meets ectoderm.



**Above pectinate line**—internal hemorrhoids, adenocarcinoma.

Arterial supply from superior rectal artery (branch of IMA).

Venous drainage: superior rectal vein → inferior mesenteric vein → splenic vein → portal vein.

Internal hemorrhoids receive visceral innervation and are therefore **not painful**. Lymphatic drainage to internal iliac lymph nodes.

**Below pectinate line**—external hemorrhoids, anal fissures, squamous cell carcinoma.

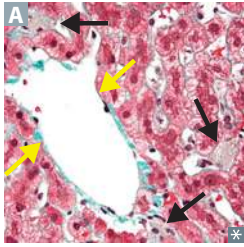
Arterial supply from inferior rectal artery (branch of internal pudendal artery).

Venous drainage: inferior rectal vein → internal pudendal vein → internal iliac vein → common iliac vein → IVC.

External hemorrhoids receive somatic innervation (inferior rectal branch of pudendal nerve) and are therefore **painful** if thrombosed. Lymphatic drainage to superficial inguinal nodes.

**Anal fissure**—tear in the anal mucosa below the Pectinate line. **P**ain while **P**ooping; blood on toilet **P**aper. Located **P**osteriorly because this area is **P**oorly **P**erfused. Associated with low-fiber diets and constipation.

### Liver tissue architecture



Apical surface of hepatocytes faces bile canaliculi. Basolateral surface faces sinusoids. Kupffer cells, which are specialized macrophages, form the lining of these sinusoids (black arrows in **A**; 2 yellow arrows show hepatic venule).

Hepatic stellate (Ito) cells in space of Disse store vitamin A (when quiescent) and produce extracellular matrix (when activated).

Zone I—periportal zone:

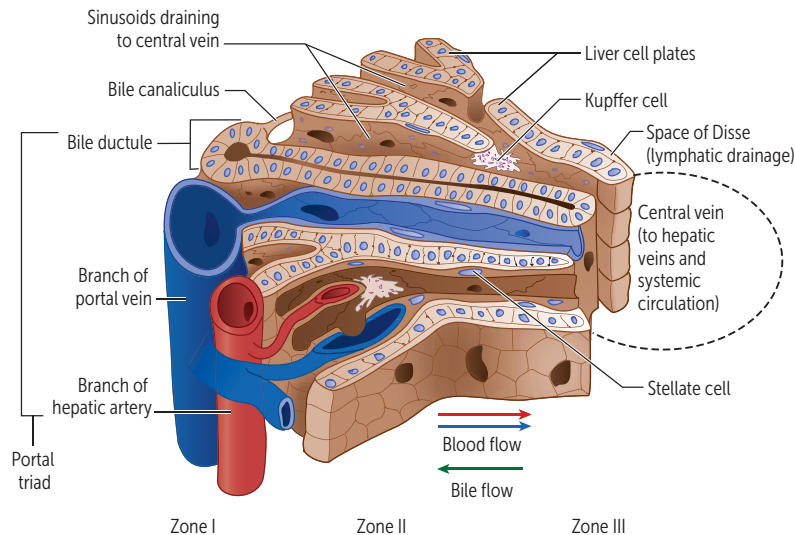
- Affected 1st by viral hepatitis
- Ingested toxins (eg, cocaine)

Zone II—intermediate zone:

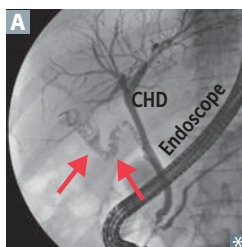
- Yellow fever

Zone III—pericentral vein (centrilobular) zone:

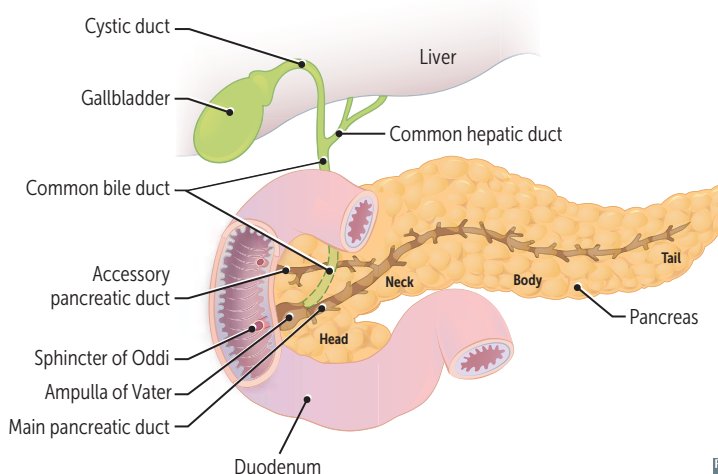
- Affected 1st by ischemia
- Contains cytochrome P-450 system
- Most sensitive to metabolic toxins
- Site of alcoholic hepatitis



### Biliary structures

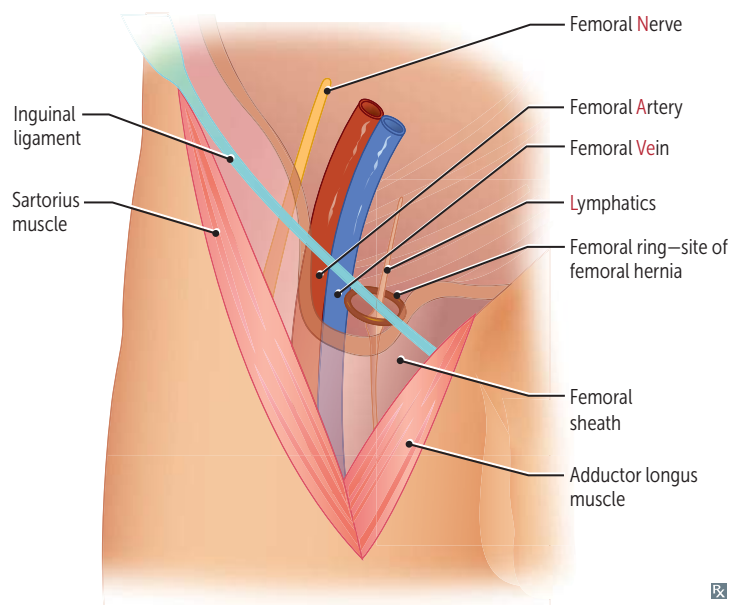


Gallstones (filling defects in gallbladder and cystic duct, red arrows in **A**) that reach the confluence of the common bile and pancreatic ducts at the ampulla of Vater can block both the common bile and pancreatic ducts (double duct sign), causing both cholangitis and pancreatitis, respectively. Tumors that arise in head of pancreas (usually ductal adenocarcinoma) can cause obstruction of common bile duct → enlarged gallbladder with painless jaundice (Courvoisier sign).

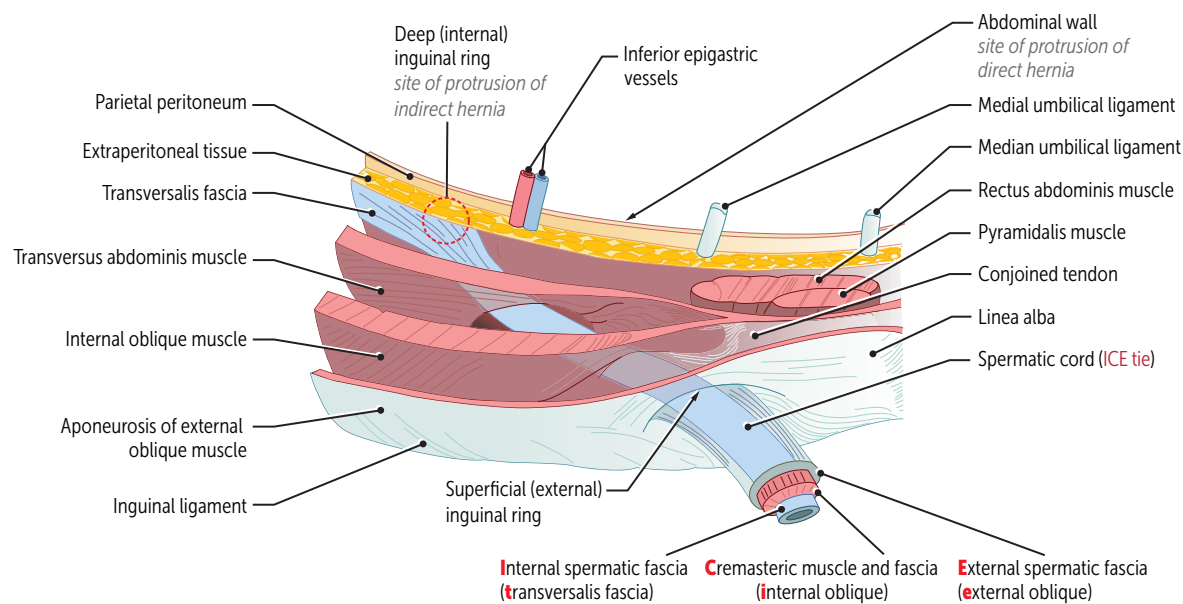


### Femoral region

|                         |                                                                                                                                          |                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| ORGANIZATION            | <b>Lateral to medial: Nerve-Artery-Vein-Lymphatics.</b>                                                                                  | You go from <b>lateral to medial</b> to find your <b>NAVeL</b> . |
| <b>Femoral triangle</b> | Contains femoral nerve, artery, vein.                                                                                                    | <b>Venous</b> near the <b>penis</b> .                            |
| <b>Femoral sheath</b>   | Fascial tube 3–4 cm below inguinal ligament. Contains femoral vein, artery, and canal (deep inguinal lymph nodes) but not femoral nerve. |                                                                  |



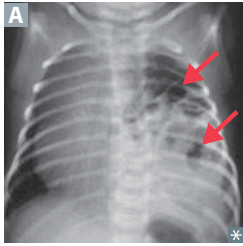
## Inguinal canal



## Hernias

A protrusion of peritoneum through an opening, usually at a site of weakness. Contents may be at risk for incarceration (not reducible back into abdomen/pelvis) and strangulation (ischemia and necrosis). Complicated hernias can present with tenderness, erythema, fever.

### Diaphragmatic hernia



Abdominal structures enter the thorax **A**; may occur due to congenital defect of pleuroperitoneal membrane, or as a result of trauma. Commonly occurs on left side due to relative protection of right hemidiaphragm by liver.

Most commonly a **hiatal hernia**, in which stomach herniates upward through the esophageal hiatus of the diaphragm.

**Sliding hiatal hernia** is most common.

Gastroesophageal junction is displaced upward; “hourglass stomach.”

**Paraesophageal hernia**—gastroesophageal junction is usually normal. Fundus protrudes into the thorax.

### Indirect inguinal hernia



Goes through the **internal** (deep) inguinal ring, external (superficial) inguinal ring, and **into** the scrotum. Enters internal inguinal ring lateral to inferior epigastric vessels. Occurs in **infants** owing to failure of processus vaginalis to close (can form hydrocele). Much more common in males **B**.

An indirect inguinal hernia follows the path of descent of the testes. Covered by all 3 layers of spermatic fascia.

### Direct inguinal hernia

Protrudes through the inguinal (Hesselbach) triangle. Bulges directly through parietal peritoneum medial to the inferior epigastric vessels but lateral to the rectus abdominis. Goes through the external (superficial) inguinal ring only. Covered by external spermatic fascia. Usually in older men, due to an acquired weakness in the transversalis fascia.

**MDs don't LIe:**

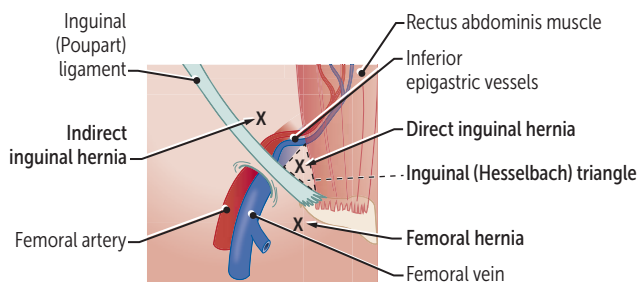
**Medial** to inferior epigastric vessels = **D**irect hernia.

**Lateral** to inferior epigastric vessels = **I**ndirect hernia.

### Femoral hernia

Protrudes below inguinal ligament through femoral canal below and lateral to pubic tubercle. More common in **females**, but overall inguinal hernias are the most common.

More likely to present with incarceration or strangulation than inguinal hernias.



Inguinal (Hesselbach) triangle:

- Inferior epigastric vessels
- Lateral border of rectus abdominis
- Inguinal ligament

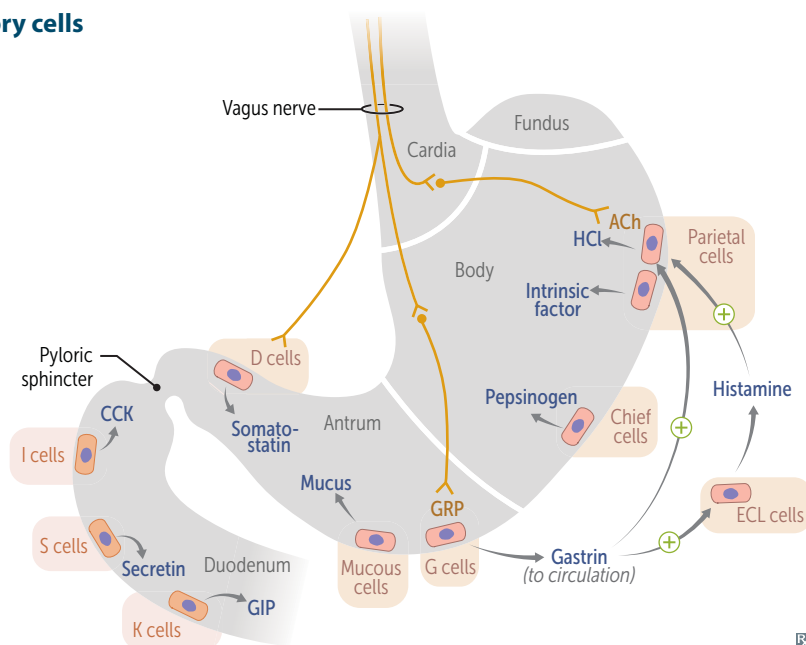
## ► GASTROINTESTINAL—PHYSIOLOGY

## Gastrointestinal regulatory substances

| REGULATORY SUBSTANCE                           | SOURCE                                                              | ACTION                                                                                                                                                              | REGULATION                                                                                                                                               | NOTES                                                                                                                                                                   |
|------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gastrin</b>                                 | G cells (antrum of stomach, duodenum)                               | ↑ gastric H <sup>+</sup> secretion<br>↑ growth of gastric mucosa<br>↑ gastric motility                                                                              | ↑ by stomach distention/<br>alkalinization,<br>amino acids,<br>peptides, vagal<br>stimulation via<br>gastrin-releasing<br>peptide (GRP)<br>↓ by pH < 1.5 | ↑ by chronic PPI use.<br>↑ in chronic atrophic gastritis<br>(eg, <i>H pylori</i> ).<br>↑↑ in Zollinger-Ellison<br>syndrome (gastrinoma).                                |
| <b>Somatostatin</b>                            | D cells (pancreatic islets, GI mucosa)                              | ↓ gastric acid and<br>pepsinogen secretion<br>↓ pancreatic and small<br>intestine fluid secretion<br>↓ gallbladder contraction<br>↓ insulin and glucagon<br>release | ↑ by acid<br>↓ by vagal<br>stimulation                                                                                                                   | Inhibits secretion of various hormones (encourages <b>somato-stasis</b> ). Octreotide is an analog used to treat acromegaly, carcinoid syndrome, and variceal bleeding. |
| <b>Cholecystokinin</b>                         | I cells (duodenum, jejunum)                                         | ↑ pancreatic secretion<br>↑ gallbladder contraction<br>↓ gastric emptying<br>↑ sphincter of Oddi<br>relaxation                                                      | ↑ by fatty acids, amino acids                                                                                                                            | Acts on neural muscarinic pathways to cause pancreatic secretion.                                                                                                       |
| <b>Secretin</b>                                | S cells (duodenum)                                                  | ↑ pancreatic HCO <sub>3</sub> <sup>−</sup><br>secretion<br>↓ gastric acid secretion<br>↑ bile secretion                                                             | ↑ by acid, fatty acids in lumen of duodenum                                                                                                              | ↑ HCO <sub>3</sub> <sup>−</sup> neutralizes gastric acid in duodenum, allowing pancreatic enzymes to function.                                                          |
| <b>Glucose-dependent insulintropic peptide</b> | K cells (duodenum, jejunum)                                         | Exocrine:<br>↓ gastric H <sup>+</sup> secretion<br>Endocrine:<br>↑ insulin release                                                                                  | ↑ by fatty acids, amino acids, oral glucose                                                                                                              | Also known as gastric inhibitory peptide (GIP). Oral glucose load leads to ↑ insulin compared to IV equivalent due to GIP secretion.                                    |
| <b>Motilin</b>                                 | Small intestine                                                     | Produces migrating motor complexes (MMCs)                                                                                                                           | ↑ in fasting state                                                                                                                                       | Motilin receptor agonists (eg, erythromycin) are used to stimulate intestinal peristalsis.                                                                              |
| <b>Vasoactive intestinal polypeptide</b>       | Parasympathetic ganglia in sphincters, gallbladder, small intestine | ↑ intestinal water and<br>electrolyte secretion<br>↑ relaxation of intestinal<br>smooth muscle and<br>sphincters                                                    | ↑ by distention<br>and vagal<br>stimulation<br>↓ by adrenergic<br>input                                                                                  | <b>VIPoma</b> —non- $\alpha$ , non- $\beta$ islet cell pancreatic tumor that secretes VIP. <b>Watery Diarrhea, Hypokalemia, and Achlorhydria (WDHA syndrome)</b> .      |
| <b>Nitric oxide</b>                            |                                                                     | ↑ smooth muscle relaxation, including lower esophageal sphincter (LES)                                                                                              |                                                                                                                                                          | Loss of NO secretion is implicated in ↑ LES tone of achalasia.                                                                                                          |
| <b>Ghrelin</b>                                 | Stomach                                                             | ↑ appetite                                                                                                                                                          | ↑ in fasting state<br>↓ by food                                                                                                                          | ↑ in Prader-Willi syndrome.<br>↓ after gastric bypass surgery.                                                                                                          |

**Gastrointestinal secretory products**

| PRODUCT                 | SOURCE                                                                                     | ACTION                                                                                           | REGULATION                                                                      | NOTES                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Intrinsic factor</b> | Parietal cells (stomach)                                                                   | Vitamin B <sub>12</sub> -binding protein (required for B <sub>12</sub> uptake in terminal ileum) |                                                                                 | Autoimmune destruction of parietal cells → chronic gastritis and pernicious anemia.       |
| <b>Gastric acid</b>     | Parietal cells (stomach)                                                                   | ↓ stomach pH                                                                                     | ↑ by histamine, ACh, gastrin<br>↓ by somatostatin, GIP, prostaglandin, secretin |                                                                                           |
| <b>Pepsin</b>           | Chief cells (stomach)                                                                      | Protein digestion                                                                                | ↑ by vagal stimulation, local acid                                              | Pepsinogen (inactive) is converted to pepsin (active) in the presence of H <sup>+</sup> . |
| <b>Bicarbonate</b>      | Mucosal cells (stomach, duodenum, salivary glands, pancreas) and Brunner glands (duodenum) | Neutralizes acid                                                                                 | ↑ by pancreatic and biliary secretion with secretin                             | Trapped in mucus that covers the gastric epithelium.                                      |

**Locations of gastrointestinal secretory cells**

Gastrin ↑ acid secretion primarily through its effects on enterochromaffin-like (ECL) cells (leading to histamine release) rather than through its direct effect on parietal cells.

**Pancreatic secretions** Isotonic fluid; low flow → high  $\text{Cl}^-$ , high flow → high  $\text{HCO}_3^-$ .

| ENZYME                             | ROLE                                                                                                                                                                 | NOTES                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b><math>\alpha</math>-amylase</b> | Starch digestion                                                                                                                                                     | Secreted in active form                                                                                      |
| <b>Lipases</b>                     | Fat digestion                                                                                                                                                        |                                                                                                              |
| <b>Proteases</b>                   | Protein digestion                                                                                                                                                    | Includes trypsin, chymotrypsin, elastase, carboxypeptidases<br>Secreted as proenzymes also known as zymogens |
| <b>Trypsinogen</b>                 | Converted to active enzyme trypsin<br>→ activation of other proenzymes and cleaving of additional trypsinogen molecules into active trypsin (positive feedback loop) | Converted to trypsin by enterokinase/enteropeptidase, a brush-border enzyme on duodenal and jejunal mucosa   |

**Carbohydrate absorption** Only monosaccharides (glucose, galactose, fructose) are absorbed by enterocytes. Glucose and galactose are taken up by SGLT1 ( $\text{Na}^+$  dependent). Fructose is taken up by facilitated diffusion by GLUT5. All are transported to blood by GLUT2.  
D-xylose absorption test: distinguishes GI mucosal damage from other causes of malabsorption.

#### Vitamin/mineral absorption

|                                   |                                                                              |                                                                              |
|-----------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Iron</b>                       | Absorbed as $\text{Fe}^{2+}$ in duodenum.                                    | <b>Iron Fist, Bro</b>                                                        |
| <b>Folate</b>                     | Absorbed in small bowel.                                                     | Clinically relevant in patients with small bowel disease or after resection. |
| <b><math>\text{B}_{12}</math></b> | Absorbed in terminal ileum along with bile salts, requires intrinsic factor. |                                                                              |

#### Peyer patches



Unencapsulated lymphoid tissue **A** found in lamina propria and submucosa of ileum. Contain specialized M cells that sample and present antigens to immune cells.  
B cells stimulated in germinal centers of Peyer patches differentiate into IgA-secreting plasma cells, which ultimately reside in lamina propria. IgA receives protective secretory component and is then transported across the epithelium to the gut to deal with intraluminal antigen.

Think of **IgA**, the **I**ntra-gut **A**ntibody. And always say “secretory IgA.”

#### Bile

Composed of bile salts (bile acids conjugated to glycine or taurine, making them water soluble), phospholipids, cholesterol, bilirubin, water, and ions. Cholesterol 7 $\alpha$ -hydroxylase catalyzes rate-limiting step of bile acid synthesis.

Functions:

- Digestion and absorption of lipids and fat-soluble vitamins
- Cholesterol excretion (body's 1 $^\circ$  means of eliminating cholesterol)
- Antimicrobial activity (via membrane disruption)

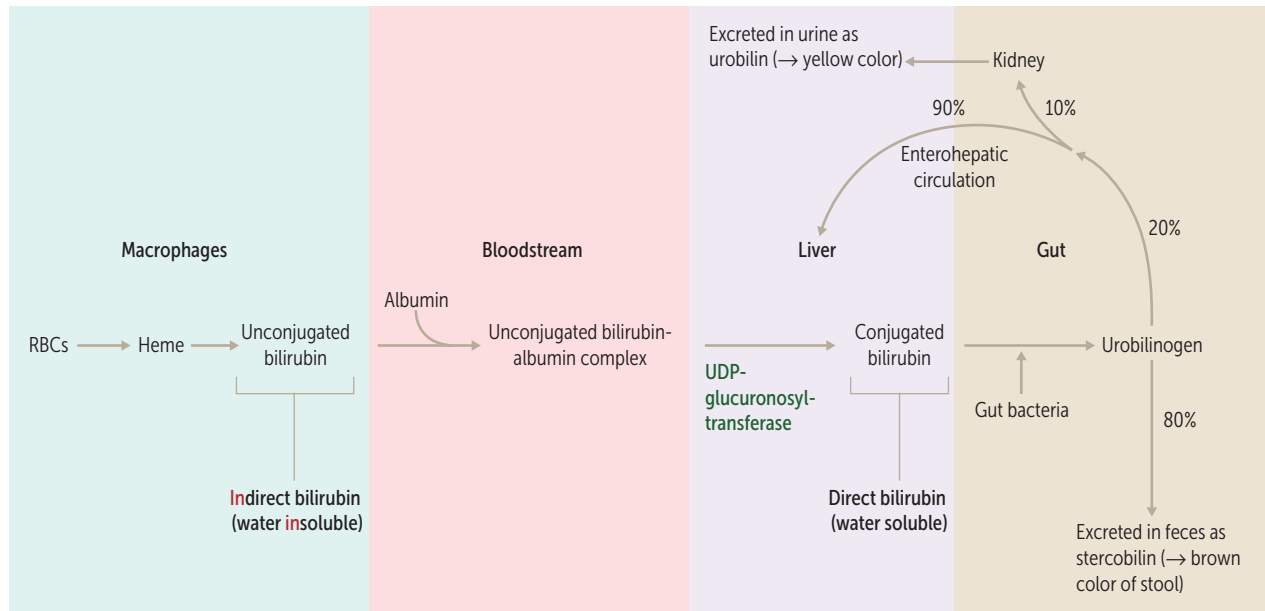


## Bilirubin

Heme is metabolized by heme oxygenase to biliverdin, which is subsequently reduced to bilirubin. Unconjugated bilirubin is removed from blood by liver, conjugated with glucuronate, and excreted in bile.

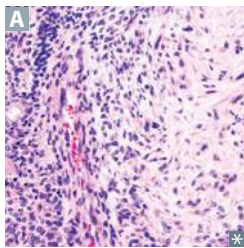
Direct bilirubin—conjugated with glucuronic acid; water soluble.

Indirect bilirubin—unconjugated; water **ins**oluble.



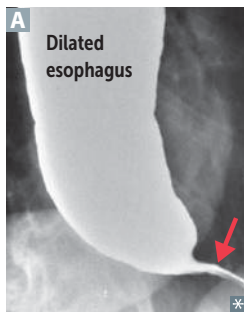
## ► GASTROINTESTINAL—PATHOLOGY

### Salivary gland tumors



Most commonly benign and in parotid gland. Tumors in smaller glands more likely malignant. Typically present as painless mass/swelling. Facial pain or paralysis suggests malignant involvement of CN VII.

- **Pleomorphic adenoma** (benign mixed tumor)—most common salivary gland tumor **A**. Composed of chondromyxoid stroma and epithelium and recurs if incompletely excised or ruptured intraoperatively.
- **Mucoepidermoid carcinoma**—most common malignant tumor, has mucinous and squamous components.
- **Warthin tumor** (papillary cystadenoma lymphomatosum)—benign cystic tumor with germinal centers. Typically found in smokers. Bilateral in 10%; malignant in 10%.

**Achalasia**

Failure of LES to relax due to loss of myenteric (Auerbach) plexus → loss of postganglionic inhibitory neurons (that contain NO and VIP). High LES resting pressure and uncoordinated or absent peristalsis → progressive dysphagia to solids and liquids (vs obstruction—solids only). Barium swallow shows dilated esophagus with an area of distal stenosis. Associated with ↑ risk of esophageal cancer.

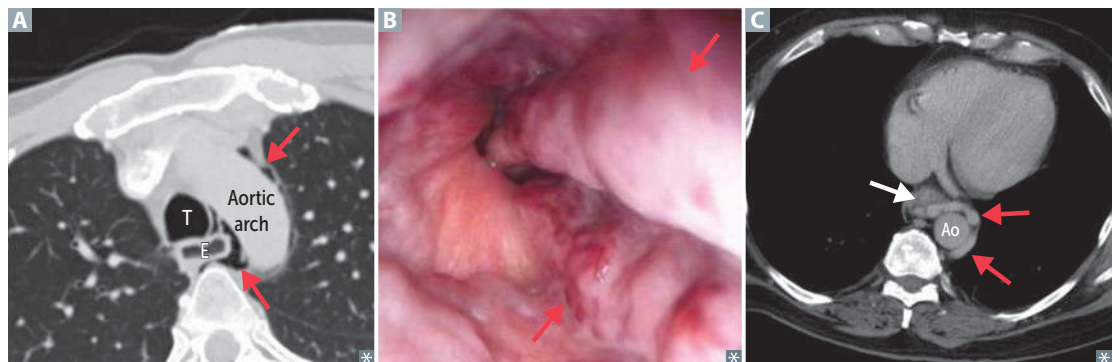
*Achalasia* = absence of relaxation.

“Bird’s beak” on barium swallow **A**.

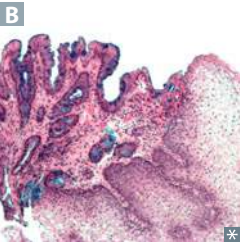
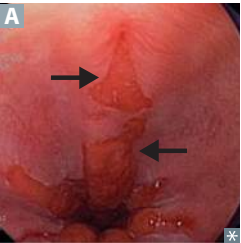
2° achalasia may arise from Chagas disease (*T cruzi* infection) or extraesophageal malignancies (mass effect or paraneoplastic).

**Esophageal pathologies**

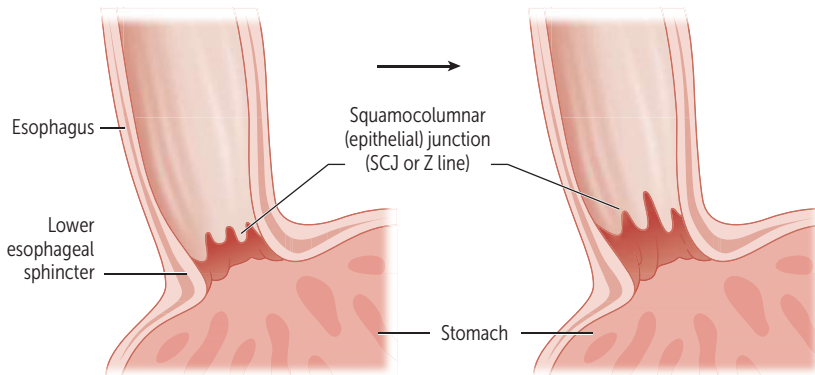
|                                           |                                                                                                                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Boerhaave syndrome</b>                 | Transmural, usually distal esophageal rupture with pneumomediastinum (arrows in <b>A</b> ) due to violent retching. Subcutaneous emphysema may be due to dissecting air (crepitus may be felt in the neck region or chest wall). Surgical emergency.              |
| <b>Eosinophilic esophagitis</b>           | Infiltration of eosinophils in the esophagus often in atopic patients. Food allergens → dysphagia, food impaction. Esophageal rings and linear furrows often seen on endoscopy. Unresponsive to GERD therapy.                                                     |
| <b>Esophageal strictures</b>              | Associated with caustic ingestion and acid reflux.                                                                                                                                                                                                                |
| <b>Esophageal varices</b>                 | Dilated submucosal veins (red arrows) <b>B C</b> in lower 1/3 of esophagus (white arrow) 2° to portal hypertension. Common in cirrhotics, may be source of life-threatening hematemesis.                                                                          |
| <b>Esophagitis</b>                        | Associated with reflux, infection in immunocompromised ( <i>Candida</i> : white pseudomembrane; HSV-1: punched-out ulcers; CMV: linear ulcers), caustic ingestion, or pill esophagitis (eg, bisphosphonates, tetracycline, NSAIDs, iron, and potassium chloride). |
| <b>Gastroesophageal reflux disease</b>    | Commonly presents as heartburn, regurgitation, dysphagia. May also present as chronic cough, hoarseness (laryngopharyngeal reflux). Associated with asthma. Transient decreases in LES tone.                                                                      |
| <b>Mallory-Weiss syndrome</b>             | Partial-thickness mucosal lacerations at gastroesophageal junction due to severe vomiting. Often presents with hematemesis and may be misdiagnosed as ruptured esophageal varices. Usually found in alcoholics and bulimics.                                      |
| <b>Plummer-Vinson syndrome</b>            | Triad of <b>D</b> ysphagia, <b>I</b> ron deficiency anemia, and <b>E</b> sophageal webs. May be associated with glossitis. Increased risk of esophageal squamous cell carcinoma (“ <b>Plumbers</b> ” <b>D</b> IE).                                                |
| <b>Scleroderma esophageal dysmotility</b> | Esophageal smooth muscle atrophy → ↓ LES pressure and dysmotility → acid reflux and dysphagia → stricture, Barrett esophagus, and aspiration. Part of CREST syndrome.                                                                                             |



**Barrett esophagus**



Specialized intestinal metaplasia **A**—replacement of nonkeratinized stratified squamous epithelium with intestinal epithelium (nonciliated columnar with goblet cells [stained blue in **B**]) in distal esophagus. Due to chronic gastroesophageal reflux (GERD). Associated with ↑ risk of esophageal adenocarcinoma.



**Esophageal cancer**

Typically presents with progressive dysphagia (first solids, then liquids) and weight loss; poor prognosis.

| CANCER                  | PART OF ESOPHAGUS AFFECTED | RISK FACTORS                                                 | PREVALENCE             |
|-------------------------|----------------------------|--------------------------------------------------------------|------------------------|
| Squamous cell carcinoma | Upper 2/3                  | Alcohol, hot liquids, caustic strictures, smoking, achalasia | More common worldwide  |
| Adenocarcinoma          | Lower 1/3                  | Chronic GERD, Barrett esophagus, obesity, smoking, achalasia | More common in America |

**Gastritis****Acute gastritis**

Erosions can be caused by:

- NSAIDs—↓ PGE<sub>2</sub> → ↓ gastric mucosa protection
- Burns (Curling ulcer)—hypovolemia → mucosal ischemia
- Brain injury (Cushing ulcer)—↑ vagal stimulation → ↑ ACh → ↑ H<sup>+</sup> production

Especially common among alcoholics and patients taking daily NSAIDs (eg, patients with rheumatoid arthritis).

**Burned** by the **Curling** iron.

Always **Cushion** the **brain**.

**Chronic gastritis**

Mucosal inflammation, often leading to atrophy (hypochlorhydria → hypergastrinemia) and intestinal G-cell metaplasia (↑ risk of gastric cancers).

*H pylori*

Most common. ↑ risk of peptic ulcer disease, MALT lymphoma.

Affects antrum first and spreads to body of stomach.

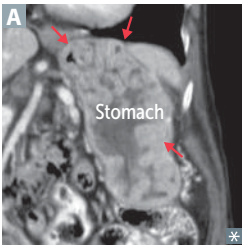
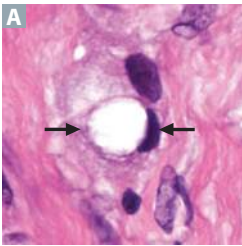
## Autoimmune

Autoantibodies to parietal cells and intrinsic factor. ↑ risk of pernicious anemia.

Affects body/fundus of stomach.

**Ménétrier disease**

Hyperplasia of gastric mucosa → hypertrophied rugae (look like brain gyri **A**), excess mucus production with resultant protein loss and parietal cell atrophy with ↓ acid production. Precancerous.

**Gastric cancer**

Most commonly gastric adenocarcinoma; lymphoma, GI stromal tumor, carcinoid (rare). Early aggressive local spread with node/liver metastases. Often presents late, with weight loss, early satiety, and in some cases acanthosis nigricans or Leser-Trélat sign.

- **Intestinal**—associated with *H pylori*, dietary nitrosamines (smoked foods), tobacco smoking, achlorhydria, chronic gastritis. Commonly on lesser curvature; looks like ulcer with raised margins.
- **Diffuse**—not associated with *H pylori*; signet ring cells (mucin-filled cells with peripheral nuclei) **A**; stomach wall grossly thickened and leathery (linitis plastica).

**Virchow node**—involvement of left supraclavicular node by metastasis from stomach.

**Krukenberg tumor**—bilateral metastases to ovaries. Abundant mucin-secreting, signet ring cells.

**Sister Mary Joseph nodule**—subcutaneous periumbilical metastasis.

**Peptic ulcer disease**

|                           | <b>Gastric ulcer</b>                          | <b>Duodenal ulcer</b>                            |
|---------------------------|-----------------------------------------------|--------------------------------------------------|
| PAIN                      | Can be <b>G</b> reater with meals—weight loss | <b>D</b> ecreases with meals—weight gain         |
| <i>H PYLORI</i> INFECTION | ~ 70%                                         | ~ 90%                                            |
| MECHANISM                 | ↓ mucosal protection against gastric acid     | ↓ mucosal protection or ↑ gastric acid secretion |
| OTHER CAUSES              | NSAIDs                                        | Zollinger-Ellison syndrome                       |
| RISK OF CARCINOMA         | ↑                                             | Generally benign                                 |
| OTHER                     | Biopsy margins to rule out malignancy         | Hypertrophy of Brunner glands                    |

**Ulcer complications****Hemorrhage**

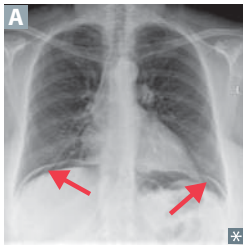
Gastric, duodenal (posterior > anterior). Most common complication.  
 Ruptured gastric ulcer on the lesser curvature of stomach → bleeding from left gastric artery.  
 An ulcer on the posterior wall of duodenum → bleeding from gastroduodenal artery.

**Obstruction**

Pyloric channel, duodenal

**Perforation**

Duodenal (anterior > posterior).  
 May see free air under diaphragm **A** with referred pain to the shoulder via irritation of phrenic nerve.



**Malabsorption syndromes**

Can cause diarrhea, steatorrhea, weight loss, weakness, vitamin and mineral deficiencies. Screen for fecal fat (eg, Sudan stain).

**Celiac disease**

Gluten-sensitive enteropathy, celiac sprue. Autoimmune-mediated intolerance of gliadin (gluten protein found in wheat) → malabsorption and steatorrhea. Associated with HLA-DQ2, HLA-DQ8, northern European descent, dermatitis herpetiformis, ↓ bone density. Findings: IgA anti-tissue transglutaminase (IgA tTG), anti-endomysial, anti-deamidated gliadin peptide antibodies; villous atrophy (arrow in **A** shows blunting), crypt hyperplasia (double arrows in **A**), and intraepithelial lymphocytosis. Moderately ↑ risk of malignancy (eg, T-cell lymphoma).

↓ mucosal absorption primarily affects distal duodenum and/or proximal jejunum.  
D-xylose test: passively absorbed in proximal small intestine; blood and urine levels ↓ with mucosa defects or bacterial overgrowth, normal in pancreatic insufficiency.  
Treatment: gluten-free diet.

**Lactose intolerance**

Lactase deficiency. Normal-appearing villi, except when 2° to injury at tips of villi (eg, viral enteritis). Osmotic diarrhea with ↓ stool pH (colonic bacteria ferment lactose).

Lactose hydrogen breath test: ⊕ for lactose malabsorption if postlactose breath hydrogen value rises > 20 ppm compared with baseline.

**Pancreatic insufficiency**

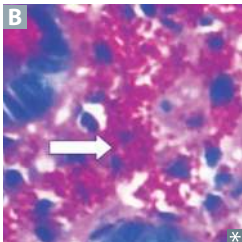
Due to chronic pancreatitis, cystic fibrosis, obstructing cancer. Causes malabsorption of fat and fat-soluble vitamins (A, D, E, K) as well as vitamin B<sub>12</sub>.

↓ duodenal pH (bicarbonate) and fecal elastase.

**Tropical sprue**

Similar findings as celiac sprue (affects small bowel), but responds to antibiotics. Cause is unknown, but seen in residents of or recent visitors to tropics.

↓ mucosal absorption affecting duodenum and jejunum but can involve ileum with time. Associated with megaloblastic anemia due to folate deficiency and, later, B<sub>12</sub> deficiency.

**Whipple disease**

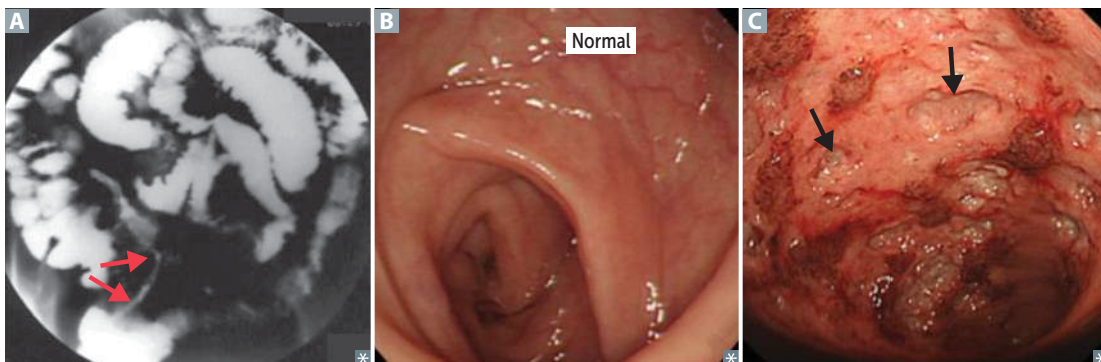
Infection with *Tropheryma whipplei* (intracellular gram ⊕); PAS ⊕ **foamy** macrophages in intestinal lamina propria **B**, mesenteric nodes. **C**ardiac symptoms, **A**rthralgias, and **N**eurologic symptoms are common. Diarrhea/steatorrhea occur later in disease course. Most common in older men.

**Foamy Whipped** cream in a **CAN**.



## Inflammatory bowel disease

|                                | Crohn disease                                                                                                                                                                                                                                                                                       | Ulcerative colitis                                                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LOCATION                       | Any portion of the GI tract, usually the terminal ileum and colon. <b>Skip</b> lesions, <b>rectal</b> sparing.                                                                                                                                                                                      | Colitis = colon inflammation. Continuous colonic lesions, always with rectal involvement.                                                                                                                                                                    |
| GROSS MORPHOLOGY               | Transmural inflammation → fistulas.<br><b>Cobblestone</b> mucosa, creeping <b>fat</b> , bowel wall thickening (“string sign” on barium swallow x-ray <b>A</b> ), linear ulcers, fissures.                                                                                                           | Mucosal and submucosal inflammation only.<br>Friable mucosa with superficial and/or deep ulcerations (compare normal <b>B</b> with diseased <b>C</b> ). Loss of haustra → “lead pipe” appearance on imaging.                                                 |
| MICROSCOPIC MORPHOLOGY         | Noncaseating <b>granulomas</b> and lymphoid aggregates. Th1 mediated.                                                                                                                                                                                                                               | Crypt abscesses and ulcers, bleeding, no granulomas. Th2 mediated.                                                                                                                                                                                           |
| COMPLICATIONS                  | Malabsorption/malnutrition, colorectal cancer (↑ risk with pancolitis).<br>Fistulas (eg, enterovesical fistulae, which can cause recurrent UTI and pneumaturia), phlegmon/abscess, strictures (causing obstruction), perianal disease.                                                              | Fulminant colitis, toxic megacolon, perforation.                                                                                                                                                                                                             |
| INTESTINAL MANIFESTATION       | Diarrhea that may or may not be bloody.                                                                                                                                                                                                                                                             | Bloody diarrhea.                                                                                                                                                                                                                                             |
| EXTRAINTESTINAL MANIFESTATIONS | Rash (pyoderma gangrenosum, erythema nodosum), eye inflammation (episcleritis, uveitis), oral ulcerations (aphthous stomatitis), arthritis (peripheral, spondylitis).<br>Kidney stones (usually calcium oxalate), gallstones. May be ⊕ for anti- <i>Saccharomyces cerevisiae</i> antibodies (ASCA). | 1° sclerosing cholangitis. Associated with p-ANCA.                                                                                                                                                                                                           |
| TREATMENT                      | Corticosteroids, azathioprine, antibiotics (eg, ciprofloxacin, metronidazole), infliximab, adalimumab.                                                                                                                                                                                              | 5-aminosalicylic preparations (eg, mesalamine), 6-mercaptopurine, infliximab, colectomy.                                                                                                                                                                     |
|                                | For <b>Crohn</b> , think of a <b>fat granny</b> and an old <b>crone</b> skipping down a <b>cobblestone</b> road away from the <b>wreck</b> (rectal sparing).                                                                                                                                        | Ulcerative colitis causes <b>ULCCERS</b> :<br><b>U</b> lcers<br><b>L</b> arge intestine<br><b>C</b> ontinuous, <b>C</b> olorectal carcinoma, <b>C</b> rypt abscesses<br><b>E</b> xtends proximally<br><b>R</b> ed diarrhea<br><b>S</b> clerosing cholangitis |



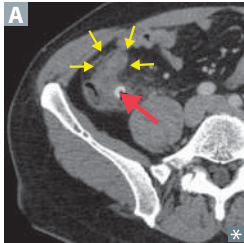
### Irritable bowel syndrome

Recurrent abdominal pain associated with  $\geq 2$  of the following:

- Related to defecation
- Change in stool frequency
- Change in form (consistency) of stool

No structural abnormalities. Most common in middle-aged women. Chronic symptoms may be diarrhea-predominant, constipation-predominant, or mixed. Pathophysiology is multifaceted.

### Appendicitis



Acute inflammation of the appendix (yellow arrows in **A**), can be due to obstruction by fecalith (red arrow in **A**) (in adults) or lymphoid hyperplasia (in children).

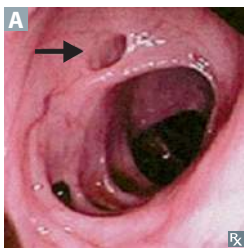
Initial diffuse periumbilical pain migrates to McBurney point ( $\frac{1}{3}$  the distance from right anterior superior iliac spine to umbilicus). Nausea, fever; may perforate  $\rightarrow$  peritonitis; may elicit psoas, obturator, and Rovsing signs, guarding and rebound tenderness on exam.

Differential: diverticulitis (elderly), ectopic pregnancy (use  $\beta$ -hCG to rule out).

Treatment: appendectomy.

### Diverticula of the GI tract

#### Diverticulum



Blind pouch **A** protruding from the alimentary tract that communicates with the lumen of the gut. Most diverticula (esophagus, stomach, duodenum, colon) are acquired and are termed “false diverticula.”

“True” diverticulum—all 3 gut wall layers outpouch (eg, Meckel).

“False” diverticulum or pseudodiverticulum—only mucosa and submucosa outpouch. Occur especially where vasa recta perforate muscularis externa.

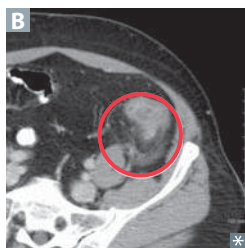
#### Diverticulosis

Many false diverticula of the colon, commonly sigmoid. Common (in  $\sim 50\%$  of people  $> 60$  years). Caused by  $\uparrow$  intraluminal pressure and focal weakness in colonic wall. Associated with low-fiber diets.

Often asymptomatic or associated with vague discomfort.

Complications include diverticular bleeding (painless hematochezia), diverticulitis.

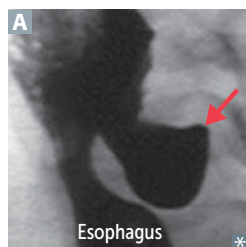
#### Diverticulitis



Inflammation of diverticula **B** classically causing LLQ pain, fever, leukocytosis. Treat with antibiotics.

Complications: abscess, fistula (colovesical fistula  $\rightarrow$  pneumaturia), obstruction (inflammatory stenosis), perforation ( $\rightarrow$  peritonitis).



**Zenker diverticulum**

Pharyngoesophageal **false** diverticulum **A**.

Esophageal dysmotility causes herniation of mucosal tissue at Killian triangle between the thyropharyngeal and cricopharyngeal parts of the inferior pharyngeal constrictor. Presenting symptoms: dysphagia, obstruction, gurgling, aspiration, foul breath, neck mass. Most common in elderly males.

**Elder MIKE** has **bad breath**.

**Elderly**

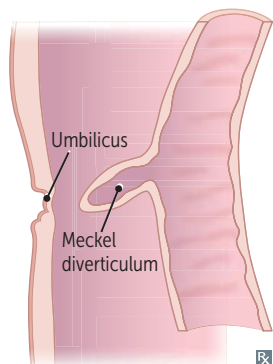
**Males**

**Inferior pharyngeal constrictor**

**Killian triangle**

**Esophageal dysmotility**

**Halitosis**

**Meckel diverticulum**

**True** diverticulum. Persistence of the vitelline duct. May contain ectopic acid-secreting gastric mucosa and/or pancreatic tissue. Most common congenital anomaly of GI tract. Can cause hematochezia/melena (less commonly), RLQ pain, intussusception, volvulus, or obstruction near terminal ileum. Contrast with omphalomesenteric cyst = cystic dilation of vitelline duct.

Diagnosis: pertechnetate study for uptake by ectopic gastric mucosa.

The rule of **2's**:

**2** times as likely in males.

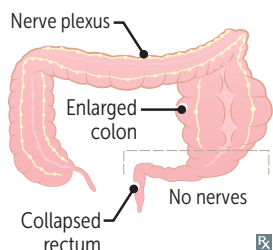
**2** inches long.

**2** feet from the ileocecal valve.

**2%** of population.

Commonly presents in first **2** years of life.

May have **2** types of epithelia (gastric/pancreatic).

**Hirschsprung disease**

Congenital megacolon characterized by lack of ganglion cells/enteric nervous plexuses (Auerbach and Meissner plexuses) in distal segment of colon. Due to failure of neural crest cell migration. Associated with mutations in *RET*.

Presents with bilious emesis, abdominal distention, and failure to pass meconium within 48 hours → chronic constipation. Normal portion of the colon proximal to the aganglionic segment is dilated, resulting in a “transition zone.”

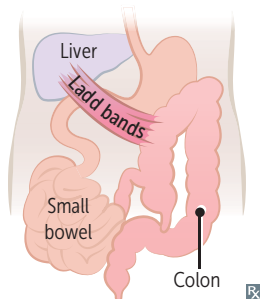
Risk ↑ with Down syndrome.

Explosive expulsion of feces (squirt sign)

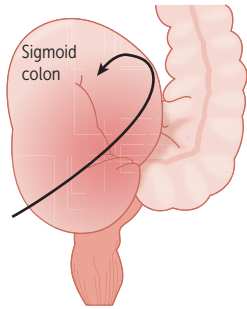
→ empty rectum on digital exam.

Diagnosed by absence of ganglionic cells on rectal suction biopsy.

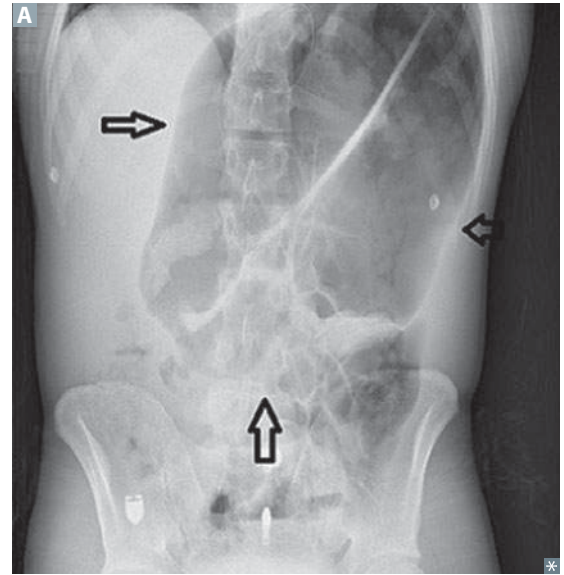
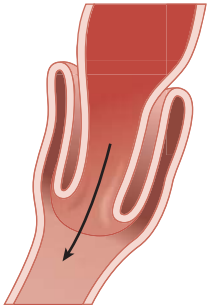
Treatment: resection.

**Malrotation**

Anomaly of midgut rotation during fetal development → improper positioning of bowel, formation of fibrous bands (Ladd bands). Can lead to volvulus, duodenal obstruction.

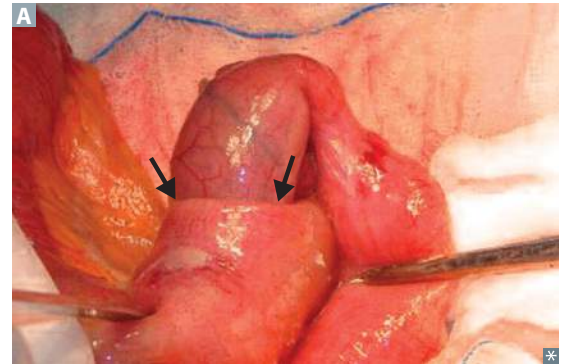
**Volvulus**

Twisting of portion of bowel around its mesentery; can lead to obstruction and infarction. Can occur throughout the GI tract. Midgut volvulus more common in infants and children. Sigmoid volvulus (coffee bean sign on x-ray **A**) more common in elderly.

**Intussusception**

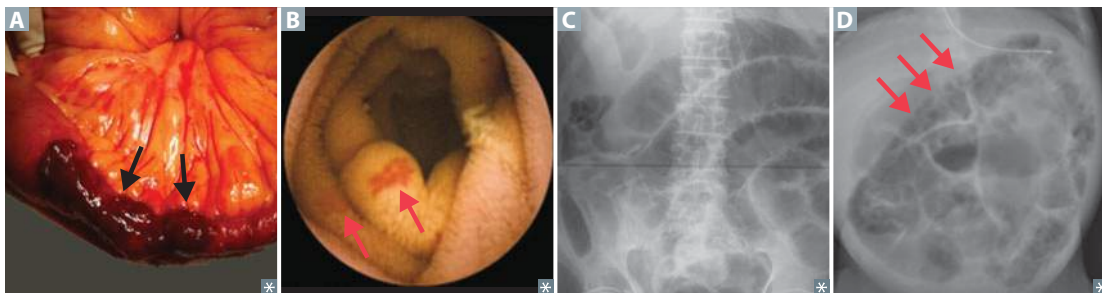
Telescoping **A** of proximal bowel segment into distal segment, commonly at ileocecal junction. Compromised blood supply → intermittent abdominal pain often with “currant jelly” stools. Unusual in adults (associated with intraluminal mass or tumor that acts as lead point that is pulled into the lumen). Most common pathologic lead point is Meckel diverticulum.

Majority of cases occur in children and are idiopathic. May be associated with recent viral infection, such as adenovirus → Peyer patch hypertrophy → lead point. Also associated with rotavirus vaccine.



**Other intestinal disorders**

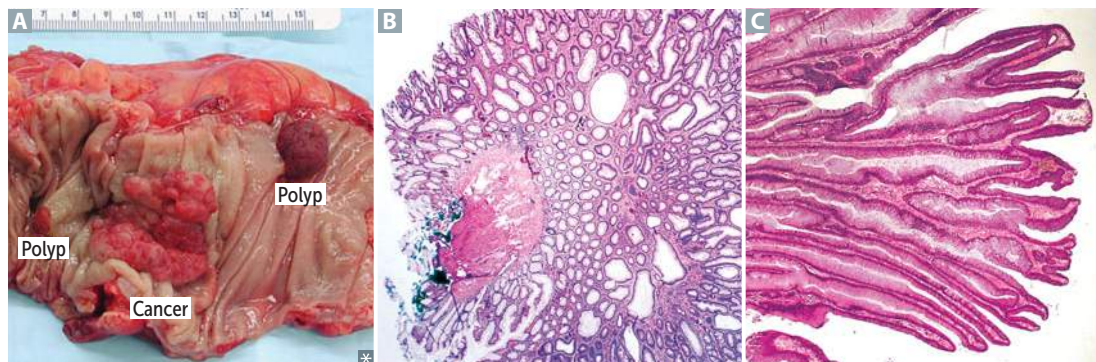
|                                    |                                                                                                                                                                                                                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acute mesenteric ischemia</b>   | Critical blockage of intestinal blood flow (often embolic occlusion of SMA) → small bowel necrosis <b>A</b> → abdominal pain out of proportion to physical findings. May see red “currant jelly” stools.                                                                                      |
| <b>Chronic mesenteric ischemia</b> | “Intestinal angina”: atherosclerosis of celiac artery, SMA, or IMA → intestinal hypoperfusion → postprandial epigastric pain → food aversion and weight loss.                                                                                                                                 |
| <b>Colonic ischemia</b>            | Reduction in intestinal blood flow causes ischemia. Crampy abdominal pain followed by hematochezia. Commonly occurs at watershed areas (splenic flexure, distal colon). Typically affects elderly. Thumbprint sign on imaging due to mucosal edema/hemorrhage.                                |
| <b>Angiodysplasia</b>              | Tortuous dilation of vessels <b>B</b> → hematochezia. Most often found in the right-sided colon. More common in older patients. Confirmed by angiography.                                                                                                                                     |
| <b>Adhesion</b>                    | Fibrous band of scar tissue; commonly forms after surgery; most common cause of small bowel obstruction <b>C</b> . Can have well-demarcated necrotic zones.                                                                                                                                   |
| <b>Ileus</b>                       | Intestinal hypomotility without obstruction → constipation and ↓ flatus; distended/tympanic abdomen with ↓ bowel sounds. Associated with abdominal surgeries, opiates, hypokalemia, sepsis. Treatment: bowel rest, electrolyte correction, cholinergic drugs (stimulate intestinal motility). |
| <b>Meconium ileus</b>              | In cystic fibrosis, meconium plug obstructs intestine, preventing stool passage at birth.                                                                                                                                                                                                     |
| <b>Necrotizing enterocolitis</b>   | Seen in premature, formula-fed infants with immature immune system. Necrosis of intestinal mucosa (primarily colonic) with possible perforation, which can lead to pneumatosis intestinalis <b>D</b> , free air in abdomen, portal venous gas.                                                |



**Colonic polyps**

Growths of tissue within the colon **A**. May be neoplastic or non-neoplastic. Grossly characterized as flat, sessile, or pedunculated (on a stalk) on the basis of protrusion into colonic lumen. Generally classified by histologic type.

| HISTOLOGIC TYPE                  | CHARACTERISTICS                                                                                                                                                                                                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generally non-neoplastic</b>  |                                                                                                                                                                                                                                                                                                                                |
| <b>Hamartomatous polyps</b>      | Solitary lesions do not have significant risk of transformation. Growths of normal colonic tissue with distorted architecture. Associated with Peutz-Jeghers syndrome and juvenile polyposis.                                                                                                                                  |
| <b>Mucosal polyps</b>            | Small, usually < 5 mm. Look similar to normal mucosa. Clinically insignificant.                                                                                                                                                                                                                                                |
| <b>Inflammatory pseudopolyps</b> | Result of mucosal erosion in inflammatory bowel disease.                                                                                                                                                                                                                                                                       |
| <b>Submucosal polyps</b>         | May include lesions such as lipomas, leiomyomas, fibromas, and others.                                                                                                                                                                                                                                                         |
| <b>Hyperplastic polyps</b>       | Generally smaller and predominantly located in the rectosigmoid region. May occasionally evolve into serrated polyps and more advanced lesions.                                                                                                                                                                                |
| <b>Malignant potential</b>       |                                                                                                                                                                                                                                                                                                                                |
| <b>Adenomatous polyps</b>        | Neoplastic, via chromosomal instability pathway with mutations in APC and KRAS. Tubular <b>B</b> histology has less malignant potential than villous <b>C</b> (“ <b>VILLOUS</b> histology is <b>VILLainOUS</b> ”); tubulovillous has intermediate malignant potential. Usually asymptomatic; may present with occult bleeding. |
| <b>Serrated polyps</b>           | Premalignant, via CpG hypermethylation phenotype pathway with microsatellite instability and mutations in BRAF. “Saw-tooth” pattern of crypts on biopsy. Up to 20% of cases of sporadic CRC.                                                                                                                                   |

**Polyposis syndromes**

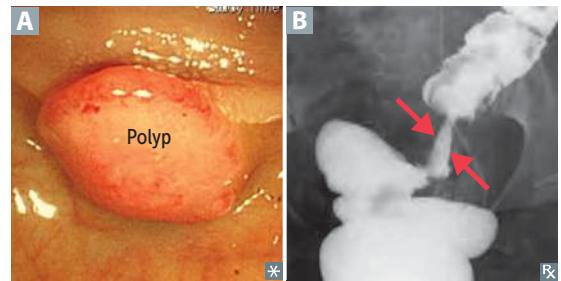
|                                       |                                                                                                                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Familial adenomatous polyposis</b> | Autosomal dominant mutation of APC tumor suppressor gene on chromosome 5q. 2-hit hypothesis. Thousands of polyps arise starting after puberty; pancolonic; always involves rectum. Prophylactic colectomy or else 100% progress to CRC.     |
| <b>Gardner syndrome</b>               | FAP + osseous and soft tissue tumors, congenital hypertrophy of retinal pigment epithelium, impacted/supernumerary teeth.                                                                                                                   |
| <b>Turcot syndrome</b>                | FAP/Lynch syndrome + malignant CNS tumor (eg, medulloblastoma, glioma). <b>Turcot</b> = <b>Turban</b> .                                                                                                                                     |
| <b>Peutz-Jeghers syndrome</b>         | Autosomal dominant syndrome featuring numerous hamartomas throughout GI tract, along with hyperpigmented mouth, lips, hands, genitalia. Associated with ↑ risk of breast and GI cancers (eg, colorectal, stomach, small bowel, pancreatic). |
| <b>Juvenile polyposis syndrome</b>    | Autosomal dominant syndrome in children (typically < 5 years old) featuring numerous hamartomatous polyps in the colon, stomach, small bowel. Associated with ↑ risk of CRC.                                                                |

**Lynch syndrome**

Previously known as hereditary nonpolyposis colorectal cancer (HNPCC). Autosomal dominant mutation of DNA mismatch repair genes with subsequent microsatellite instability. ~ 80% progress to CRC. Proximal colon is always involved. Associated with endometrial, ovarian, and skin cancers.

**Colorectal cancer**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EPIDEMIOLOGY | Most patients are > 50 years old. ~ 25% have a family history.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| RISK FACTORS | Adenomatous and serrated polyps, familial cancer syndromes, IBD, tobacco use, diet of processed meat with low fiber.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PRESENTATION | <p>Rectosigmoid &gt; ascending &gt; descending. Right side bleeds; left side obstructs.</p> <p>Ascending—exophytic mass, iron deficiency anemia, weight loss.</p> <p>Descending—infiltrating mass, partial obstruction, colicky pain, hematochezia.</p> <p>Rarely, presents with <i>S bovis</i> (<i>gallolyticus</i>) bacteremia.</p>                                                                                                                                                                                                                                                                                                                                                                                            |
| DIAGNOSIS    | <p>Iron deficiency anemia in males (especially &gt; 50 years old) and postmenopausal females raises suspicion.</p> <p>Screen low-risk patients starting at age 50 with colonoscopy <b>A</b>; alternatives include flexible sigmoidoscopy, fecal occult blood testing (FOBT), fecal immunochemical testing (FIT), and CT colonography. Patients with a first-degree relative who has colon cancer should be screened via colonoscopy at age 40, or starting 10 years prior to their relative's presentation. Patients with IBD have a distinct screening protocol.</p> <p>“Apple core” lesion seen on barium enema x-ray <b>B</b>.</p> <p>CEA tumor marker: good for monitoring recurrence, should not be used for screening.</p> |





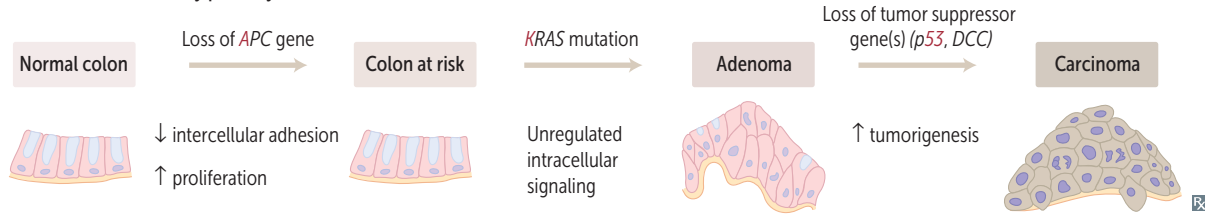
### Molecular pathogenesis of colorectal cancer

Chromosomal instability pathway: mutations in *APC* cause FAP and most sporadic CRC (via adenoma-carcinoma sequence; **(firing)** order of events is **AK-53**).

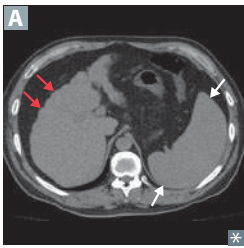
Microsatellite instability pathway: mutations or methylation of mismatch repair genes (eg, *MLH1*) cause Lynch syndrome and some sporadic CRC (via serrated polyp pathway).

Overexpression of COX-2 has been linked to colorectal cancer, NSAIDs may be chemopreventive.

#### Chromosomal instability pathway

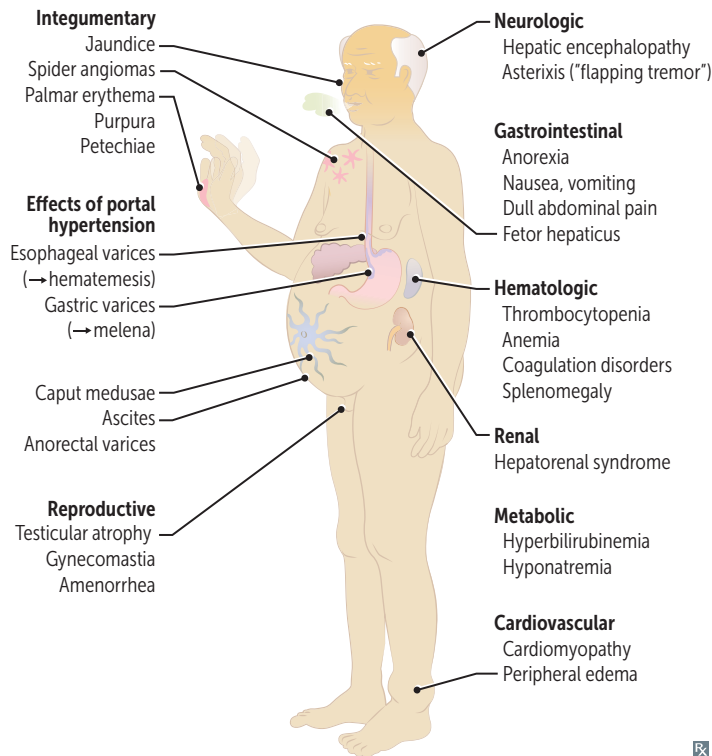


### Cirrhosis and portal hypertension



**Cirrhosis**—diffuse bridging fibrosis (via stellate cells) and regenerative nodules (red arrows in **A**; white arrows shows splenomegaly) disrupt normal architecture of liver; ↑ risk for hepatocellular carcinoma (HCC). Etiologies include alcohol (60–70% of cases in the US), nonalcoholic steatohepatitis, chronic viral hepatitis, autoimmune hepatitis, biliary disease, genetic/metabolic disorders.

**Portal hypertension**—↑ pressure in portal venous system. Etiologies include cirrhosis (most common cause in Western countries), vascular obstruction (eg, portal vein thrombosis, Budd-Chiari syndrome), schistosomiasis.



**Spontaneous bacterial peritonitis** Also known as 1° bacterial peritonitis. Common and potentially fatal bacterial infection in patients with cirrhosis and ascites. Often asymptomatic, but can cause fevers, chills, abdominal pain, ileus, or worsening encephalopathy. Commonly caused by aerobic gram  $\ominus$  organisms, especially *E coli*. Diagnosis: Paracentesis with ascitic fluid absolute neutrophil count (ANC)  $> 250$  cells/mm<sup>3</sup>.

### Serum markers of liver pathology

#### ENZYMES RELEASED IN LIVER DAMAGE

**Aspartate aminotransferase and alanine aminotransferase**  $\uparrow$  in most liver disease: ALT  $>$  AST  
 $\uparrow$  in alcoholic liver disease: AST  $>$  ALT  
 AST  $>$  ALT in nonalcoholic liver disease suggests progression to advanced fibrosis or cirrhosis

**Alkaline phosphatase**  $\uparrow$  in cholestasis (eg, biliary obstruction), infiltrative disorders, bone disease

**$\gamma$ -glutamyl transpeptidase**  $\uparrow$  in various liver and biliary diseases (just as ALP can), but not in bone disease; associated with alcohol use

#### FUNCTIONAL LIVER MARKERS

**Bilirubin**  $\uparrow$  in various liver diseases (eg, biliary obstruction, alcoholic or viral hepatitis, cirrhosis), hemolysis

**Albumin**  $\downarrow$  in advanced liver disease (marker of liver's biosynthetic function)

**Prothrombin time**  $\uparrow$  in advanced liver disease ( $\downarrow$  production of clotting factors, thereby measuring the liver's biosynthetic function)

**Platelets**  $\downarrow$  in advanced liver disease ( $\downarrow$  thrombopoietin, liver sequestration) and portal hypertension (splenomegaly/splenic sequestration)

**Reye syndrome** Rare, often fatal childhood hepatic encephalopathy. Findings: mitochondrial abnormalities, fatty liver (microvesicular fatty change), hypoglycemia, vomiting, hepatomegaly, coma. Associated with viral infection (especially VZV and influenza B) that has been treated with aspirin. Mechanism: aspirin metabolites  $\downarrow$   $\beta$ -oxidation by reversible inhibition of mitochondrial enzymes. Avoid aspirin in children, except in those with Kawasaki disease.

**Alcoholic liver disease****Hepatic steatosis**

Macrovesicular fatty change **A** that may be reversible with alcohol cessation.

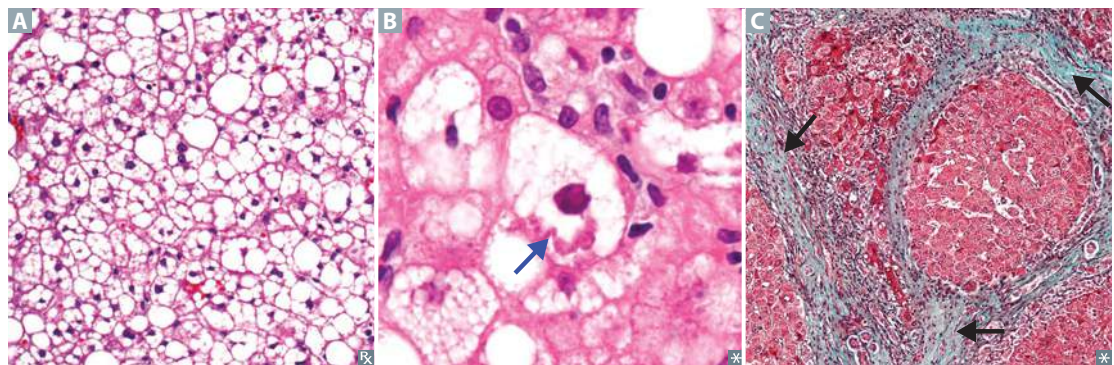
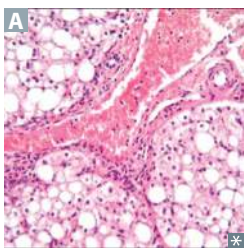
**Alcoholic hepatitis**

Requires sustained, long-term consumption. Swollen and necrotic hepatocytes with neutrophilic infiltration. Mallory bodies **B** (intracytoplasmic eosinophilic inclusions of damaged keratin filaments).

Make a to**AST** with alcohol:  
**AST** > ALT (ratio usually > 2:1).

**Alcoholic cirrhosis**

Final and usually irreversible form. Regenerative nodules surrounded by fibrous bands in response to chronic liver injury → portal hypertension and end-stage liver disease. Sclerosis around central vein (arrows in **C**) may be seen in early disease.

**Nonalcoholic fatty liver disease**

Metabolic syndrome (insulin resistance); obesity → fatty infiltration of hepatocytes **A** → cellular “ballooning” and eventual necrosis. May cause cirrhosis and HCC. Independent of alcohol use.

ALT > AST (**L**ipids)

**Hepatic encephalopathy**

Cirrhosis → portosystemic shunts → ↓ NH<sub>3</sub> metabolism → neuropsychiatric dysfunction. Reversible neuropsychiatric dysfunction ranging from disorientation/asterixis (mild) to difficult arousal or coma (severe). Triggers:

- ↑ NH<sub>3</sub> production and absorption (due to dietary protein, GI bleed, constipation, infection).
- ↓ NH<sub>3</sub> removal (due to renal failure, diuretics, bypassed hepatic blood flow post-TIPS).

Treatment: lactulose (↑ NH<sub>4</sub><sup>+</sup> generation) and rifaximin or neomycin (↓ NH<sub>3</sub> producing gut bacteria).

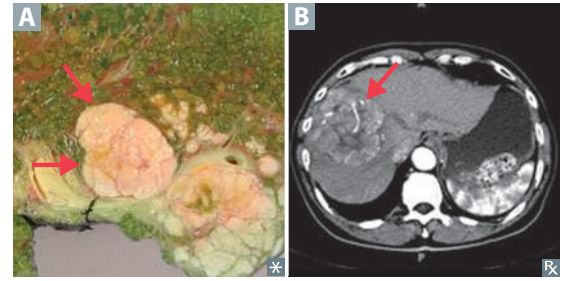


### Hepatocellular carcinoma/hepatoma

Most common 1° malignant tumor of liver in adults **A**. Associated with HBV (+/– cirrhosis) and all other causes of cirrhosis (including HCV, alcoholic and nonalcoholic fatty liver disease, autoimmune disease, hemochromatosis,  $\alpha_1$ -antitrypsin deficiency) and specific carcinogens (eg, aflatoxin from *Aspergillus*). May lead to Budd-Chiari syndrome.

Findings: jaundice, tender hepatomegaly, ascites, polycythemia, anorexia. Spreads hematogenously.

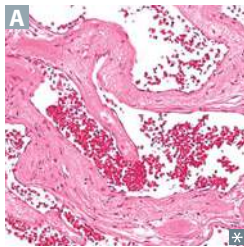
Diagnosis: ↑  $\alpha$ -fetoprotein; ultrasound or contrast CT/MRI **B**, biopsy.



### Other liver tumors

#### Cavernous hemangioma

Common, benign liver tumor **A**; typically occurs at age 30–50 years. Biopsy contraindicated because of risk of hemorrhage.



#### Hepatic adenoma

Rare, benign liver tumor, often related to oral contraceptive or anabolic steroid use; may regress spontaneously or rupture (abdominal pain and shock).

#### Angiosarcoma

Malignant tumor of endothelial origin; associated with exposure to arsenic, vinyl chloride.

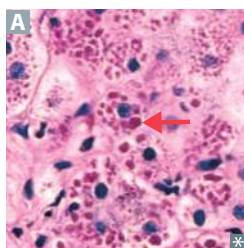
#### Metastases

GI malignancies, breast and lung cancer. Most common overall; metastases are rarely solitary.

### Budd-Chiari syndrome

Thrombosis or compression of hepatic veins with centrilobular congestion and necrosis → congestive liver disease (hepatomegaly, ascites, varices, abdominal pain, liver failure). Absence of JVD. Associated with hypercoagulable states, polycythemia vera, postpartum state, HCC. May cause nutmeg liver (mottled appearance).

### $\alpha_1$ -antitrypsin deficiency



Misfolded gene product protein aggregates in hepatocellular ER → cirrhosis with PAS ⊕ globules **A** in liver. Codominant trait. Often presents in young patients with liver damage and dyspnea without a history of smoking.

In lungs, ↓  $\alpha_1$ -antitrypsin → uninhibited elastase in alveoli → ↓ elastic tissue → panacinar emphysema.

**Jaundice**

Abnormal yellowing of the skin and/or sclera **A** due to bilirubin deposition. Hyperbilirubinemia 2° to ↑ production or ↓ disposition (impaired hepatic uptake, conjugation, excretion).

**HOT Liver**—common causes of increased levels of bilirubin:

**H**emolysis  
**O**bstuction  
**T**umor  
**L**iver disease

**Unconjugated  
 (indirect)  
 hyperbilirubinemia**

Hemolytic, physiologic (newborns), Crigler-Najjar, Gilbert syndrome.

**Conjugated (direct)  
 hyperbilirubinemia**

Biliary tract obstruction: gallstones, cholangiocarcinoma, pancreatic or liver cancer, liver fluke.

Biliary tract disease:

- 1° sclerosing cholangitis
- 1° biliary cholangitis

Excretion defect: Dubin-Johnson syndrome, Rotor syndrome.

**Mixed (direct  
 and indirect)  
 hyperbilirubinemia**

Hepatitis, cirrhosis.

**Physiologic  
 neonatal jaundice**

At birth, immature UDP-glucuronosyltransferase → unconjugated hyperbilirubinemia → jaundice/kernicterus (deposition of unconjugated, lipid-soluble bilirubin in the brain, particularly basal ganglia).

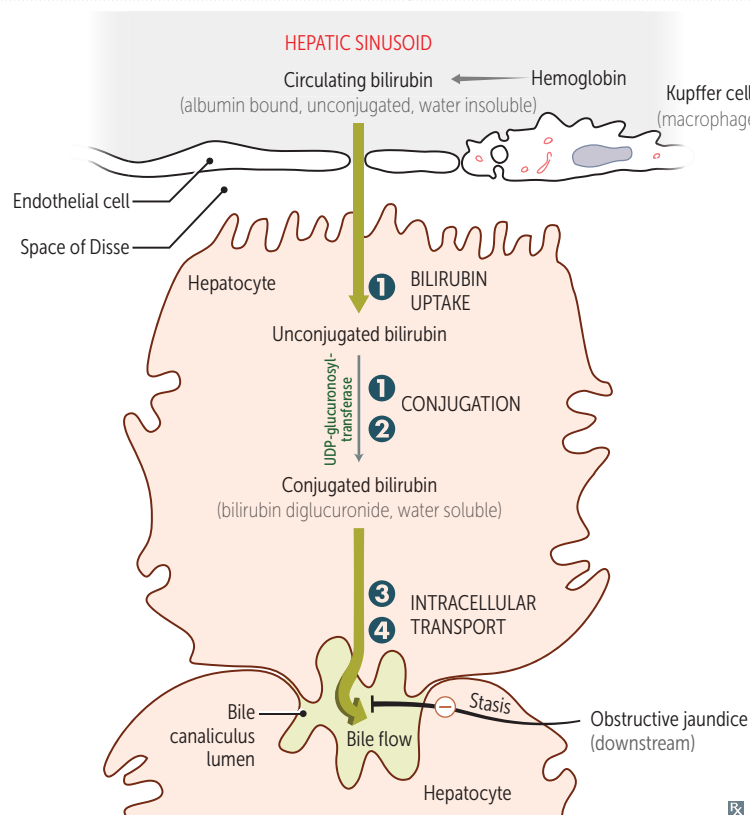
Occurs after first 24 hours of life and usually resolves without treatment in 1–2 weeks.

Treatment: phototherapy (non-UV) isomerizes unconjugated bilirubin to water-soluble form.

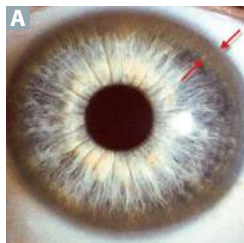
**Hereditary hyperbilirubinemias**

All autosomal recessive.

|                                          |                                                                                                                                                                                                                                                |                                                                                                                                   |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>1 Gilbert syndrome</b>                | Mildly ↓ UDP-glucuronosyltransferase conjugation and impaired bilirubin uptake. Asymptomatic or mild jaundice usually with stress, illness, or fasting. ↑ unconjugated bilirubin without overt hemolysis. Bilirubin ↑ with fasting and stress. | Relatively common, benign condition.                                                                                              |
| <b>2 Crigler-Najjar syndrome, type I</b> | Absent UDP-glucuronosyltransferase. Presents early in life; patients die within a few years. Findings: jaundice, kernicterus (bilirubin deposition in brain), ↑ unconjugated bilirubin. Treatment: plasmapheresis and phototherapy.            | Type II is less severe and responds to phenobarbital, which ↑ liver enzyme synthesis.                                             |
| <b>3 Dubin-Johnson syndrome</b>          | Conjugated hyperbilirubinemia due to defective liver excretion. Grossly black liver. Benign.                                                                                                                                                   | <b>4 Rotor syndrome</b> is similar, but milder in presentation without black liver. Due to impaired hepatic uptake and excretion. |



### Wilson disease (hepatolenticular degeneration)

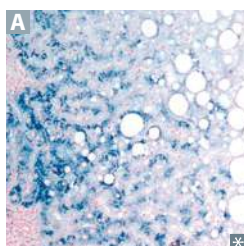


Autosomal recessive mutations in hepatocyte copper-transporting ATPase (*ATP7B* gene; chromosome 13) → ↓ copper excretion into bile and incorporation into apoceruloplasmin → ↓ serum ceruloplasmin. Copper accumulates, especially in liver, brain, cornea, kidneys; ↑ urine copper.

Presents before age 40 with liver disease (eg, hepatitis, acute liver failure, cirrhosis), neurologic disease (eg, dysarthria, dystonia, tremor, parkinsonism), psychiatric disease, Kayser-Fleischer rings (deposits in Descemet membrane of cornea) **A**, hemolytic anemia, renal disease (eg, Fanconi syndrome).

Treatment: chelation with penicillamine or trientine, oral zinc.

### Hemochromatosis



Recessive mutations in *HFE* gene (C282Y > H63D, chromosome 6, associated with HLA-A3) → abnormal iron sensing and ↑ intestinal absorption (↑ ferritin, ↑ iron, ↓ TIBC → ↑ transferrin saturation). Iron overload can also be 2° to chronic transfusion therapy (eg, β-thalassemia major). Iron accumulates, especially in liver, pancreas, skin, heart, pituitary, joints. Hemosiderin (iron) can be identified on liver MRI or biopsy with Prussian blue stain **A**.

Presents after age 40 when total body iron > 20 g; iron loss through menstruation slows progression in women. Classic triad of cirrhosis, diabetes mellitus, skin pigmentation (“bronze diabetes”). Also causes restrictive cardiomyopathy (classic) or dilated cardiomyopathy (reversible), hypogonadism, arthropathy (calcium pyrophosphate deposition; especially metacarpophalangeal joints). HCC is common cause of death.

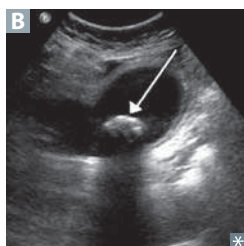
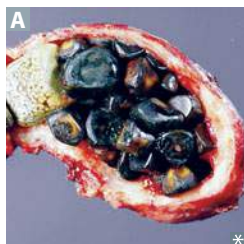
Treatment: repeated phlebotomy, chelation with deferasirox, deferoxamine, oral deferiprone.

### Biliary tract disease

May present with pruritus, jaundice, dark urine, light-colored stool, hepatosplenomegaly. Typically with cholestatic pattern of LFTs (↑ conjugated bilirubin, ↑ cholesterol, ↑ ALP).

|                                       | PATHOLOGY                                                                                                                                                                                                           | EPIDEMIOLOGY                                                                                    | ADDITIONAL FEATURES                                                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary sclerosing cholangitis</b> | Unknown cause of concentric “onion skin” bile duct fibrosis → alternating strictures and dilation with “beading” of intra- and extrahepatic bile ducts on ERCP, magnetic resonance cholangiopancreatography (MRCP). | Classically in middle-aged men with IBD.                                                        | Associated with ulcerative colitis. p-ANCA ⊕. ↑ IgM. Can lead to 2° biliary cholangitis. ↑ risk of cholangiocarcinoma and gallbladder cancer.                                 |
| <b>Primary biliary cholangitis</b>    | Autoimmune reaction → lymphocytic infiltrate + granulomas → destruction of intralobular bile ducts.                                                                                                                 | Classically in middle-aged women.                                                               | Anti-mitochondrial antibody ⊕, ↑ IgM. Associated with other autoimmune conditions (eg, Sjögren syndrome, Hashimoto thyroiditis, CREST, rheumatoid arthritis, celiac disease). |
| <b>Secondary biliary cholangitis</b>  | Extrahepatic biliary obstruction → ↑ pressure in intrahepatic ducts → injury/ fibrosis and bile stasis.                                                                                                             | Patients with known obstructive lesions (gallstones, biliary strictures, pancreatic carcinoma). | May be complicated by ascending cholangitis.                                                                                                                                  |

## Gallstones (cholelithiasis)



↑ cholesterol and/or bilirubin, ↓ bile salts, and gallbladder stasis all cause stones.

2 types of stones:

- Cholesterol stones (radiolucent with 10–20% opaque due to calcifications)—80% of stones. Associated with obesity, Crohn disease, advanced age, estrogen therapy, multiparity, rapid weight loss, Native American origin.
- Pigment stones **A** (black = radiopaque,  $\text{Ca}^{2+}$  bilirubinate, hemolysis; brown = radiolucent, infection). Associated with Crohn disease, chronic hemolysis, alcoholic cirrhosis, advanced age, biliary infections, total parenteral nutrition (TPN).

Most common complication is cholecystitis; can also cause acute pancreatitis, ascending cholangitis.

Risk factors (**4 F's**):

- F**emale
- F**at
- F**ertile (pregnant)
- F**orty

Diagnose with ultrasound **B**. Treat with elective cholecystectomy if symptomatic.

Can cause fistula between gallbladder and GI tract → air in biliary tree (pneumobilia) → passage of gallstones into intestinal tract → obstruction of ileocecal valve (gallstone ileus).

### RELATED PATHOLOGIES

### CHARACTERISTICS

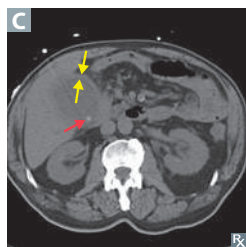
#### Biliary colic

Associated with nausea/vomiting and dull RUQ pain. Neurohormonal activation (eg, by CCK after a fatty meal) triggers contraction of gallbladder, forcing stone into cystic duct. Labs are normal, ultrasound shows cholelithiasis.

#### Choledocholithiasis

Presence of gallstone(s) in common bile duct, often leading to elevated ALP, GGT, direct bilirubin, and/or AST/ALT.

#### Cholecystitis



Acute or chronic inflammation of gallbladder usually from cholelithiasis (stone at neck of gallbladder [red arrow in **C**] with gallbladder wall thickening [yellow arrows]).

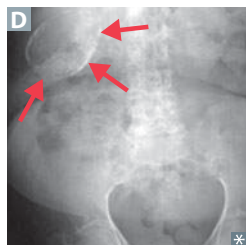
Calculous cholecystitis: most common type; due to gallstone impaction in the cystic duct resulting in inflammation; can produce 2° infection.

Acalculous cholecystitis: due to gallbladder stasis, hypoperfusion, or infection (CMV); seen in critically ill patients.

Murphy sign: inspiratory arrest on RUQ palpation due to pain. ↑ ALP if bile duct becomes involved (eg, ascending cholangitis).

Diagnose with ultrasound or cholescintigraphy (HIDA scan). Failure to visualize gallbladder on HIDA scan suggests obstruction.

#### Porcelain gallbladder



Calcified gallbladder due to chronic cholecystitis; usually found incidentally on imaging **D**.

Treatment: prophylactic cholecystectomy due to high rates of gallbladder cancer (mostly adenocarcinoma).

#### Ascending cholangitis

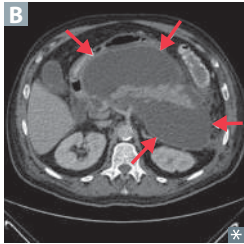
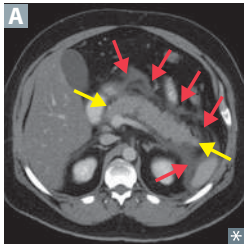
Infection of biliary tree usually due to obstruction that leads to stasis/bacterial overgrowth.

Charcot triad of cholangitis:

- Jaundice
- Fever
- RUQ pain

Reynolds pentad adds:

- Altered mental status
- Shock (hypotension)

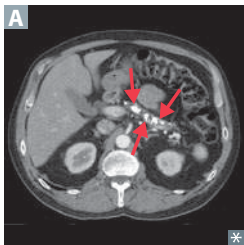
**Acute pancreatitis**

Autodigestion of pancreas by pancreatic enzymes (**A** shows pancreas [yellow arrows] surrounded by edema [red arrows]).

Causes: **I**diopathic, **G**allstones, **E**thanol, **T**rauma, **S**teroids, **M**umps, **A**utoimmune disease, **S**corpion sting, **H**ypercalcemia/**H**ypertriglyceridemia (> 1000 mg/dL), **E**RPC, **D**rugs (eg, sulfa drugs, NRTIs, protease inhibitors). **I GET SMASHED**.

Diagnosis by 2 of 3 criteria: acute epigastric pain often radiating to the back, ↑ serum amylase or lipase (more specific) to 3× upper limit of normal, or characteristic imaging findings.

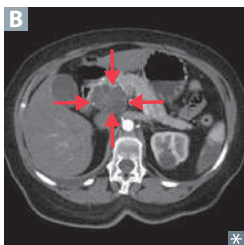
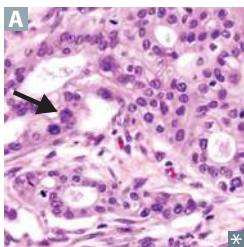
Complications: pseudocyst (**B** (lined by granulation tissue, not epithelium), necrosis, hemorrhage, infection, organ failure (ARDS, shock, renal failure), hypocalcemia (precipitation of  $\text{Ca}^{2+}$  soaps).

**Chronic pancreatitis**

Chronic inflammation, atrophy, calcification of the pancreas (**A**). Major causes are alcohol abuse and idiopathic. Complications include pancreatic insufficiency and pseudocysts.

Pancreatic insufficiency may manifest with steatorrhea, fat-soluble vitamin deficiency, diabetes mellitus.

Amylase and lipase may or may not be elevated (almost always elevated in acute pancreatitis).

**Pancreatic adenocarcinoma**

Very aggressive tumor arising from pancreatic ducts (disorganized glandular structure with cellular infiltration (**A**); often metastatic at presentation, with average survival ~ 1 year after diagnosis.

Tumors more common in pancreatic head (**B** (→ obstructive jaundice). Associated with CA 19-9 tumor marker (also CEA, less specific).

Risk factors:

- Tobacco use
- Chronic pancreatitis (especially > 20 years)
- Diabetes
- Age > 50 years
- Jewish and African-American males

Often presents with:

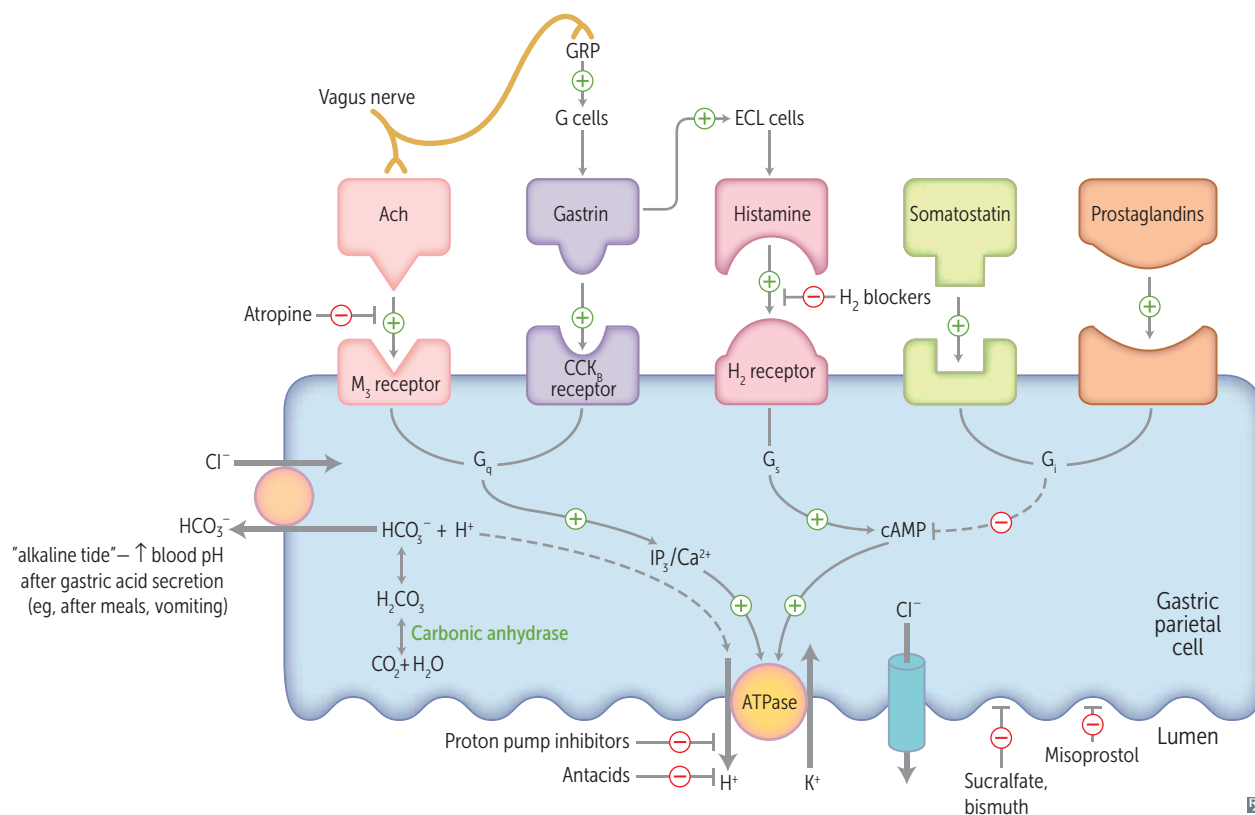
- Abdominal pain radiating to back
- Weight loss (due to malabsorption and anorexia)
- Migratory thrombophlebitis—redness and tenderness on palpation of extremities (**Trousseau syndrome**)
- Obstructive jaundice with palpable, nontender gallbladder (Courvoisier sign)

Treatment: Whipple procedure, chemotherapy, radiation therapy.



## ► GASTROINTESTINAL—PHARMACOLOGY

## Acid suppression therapy

**H<sub>2</sub> blockers**

Cimetidine, ranitidine, famotidine, nizatidine. Take H<sub>2</sub> blockers before you **dine**. Think “**table for 2**” to remember H<sub>2</sub>.

## MECHANISM

Reversible block of histamine H<sub>2</sub>-receptors → ↓ H<sup>+</sup> secretion by parietal cells.

## CLINICAL USE

Peptic ulcer, gastritis, mild esophageal reflux.

## ADVERSE EFFECTS

Cimetidine is a potent inhibitor of cytochrome P-450 (multiple drug interactions); it also has antiandrogenic effects (prolactin release, gynecomastia, impotence, ↓ libido in males); can cross blood-brain barrier (confusion, dizziness, headaches) and placenta. Both cimetidine and ranitidine ↓ renal excretion of creatinine. Other H<sub>2</sub> blockers are relatively free of these effects.

**Proton pump inhibitors** Omeprazole, lansoprazole, esomeprazole, pantoprazole, dexlansoprazole.

## MECHANISM

Irreversibly inhibit H<sup>+</sup>/K<sup>+</sup> ATPase in stomach parietal cells.

## CLINICAL USE

Peptic ulcer, gastritis, esophageal reflux, Zollinger-Ellison syndrome, component of therapy for *H. pylori*, stress ulcer prophylaxis.

## ADVERSE EFFECTS

↑ risk of *C. difficile* infection, pneumonia. ↓ serum Mg<sup>2+</sup> with long-term use.



**Antacid use**

Can affect absorption, bioavailability, or urinary excretion of other drugs by altering gastric and urinary pH or by delaying gastric emptying.  
All can cause hypokalemia.  
Overuse can also cause the following problems.

|                            |                                                                                       |                                                                    |
|----------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Aluminum hydroxide</b>  | Constipation and hypophosphatemia; proximal muscle weakness, osteodystrophy, seizures | Aluminum amount of feces.                                          |
| <b>Calcium carbonate</b>   | Hypercalcemia (milk-alkali syndrome), rebound acid ↑                                  | Can chelate and ↓ effectiveness of other drugs (eg, tetracycline). |
| <b>Magnesium hydroxide</b> | Diarrhea, hyporeflexia, hypotension, cardiac arrest                                   | Mg <sup>2+</sup> = Must go to the bathroom.                        |

**Bismuth, sucralfate**

|              |                                                                                                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MECHANISM    | Bind to ulcer base, providing physical protection and allowing HCO <sub>3</sub> <sup>-</sup> secretion to reestablish pH gradient in the mucous layer. Require acidic environment; usually not given with PPIs/H <sub>2</sub> blockers. |
| CLINICAL USE | ↑ ulcer healing, travelers' diarrhea.                                                                                                                                                                                                   |

**Misoprostol**

|                 |                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| MECHANISM       | A PGE <sub>1</sub> analog. ↑ production and secretion of gastric mucous barrier, ↓ acid production.                                               |
| CLINICAL USE    | Prevention of NSAID-induced peptic ulcers (NSAIDs block PGE <sub>1</sub> production). Also used off-label for induction of labor (ripens cervix). |
| ADVERSE EFFECTS | Diarrhea. Contraindicated in women of childbearing potential (abortifacient).                                                                     |

**Octreotide**

|                 |                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------|
| MECHANISM       | Long-acting somatostatin analog; inhibits secretion of various splanchnic vasodilatory hormones. |
| CLINICAL USE    | Acute variceal bleeds, acromegaly, VIPoma, carcinoid tumors.                                     |
| ADVERSE EFFECTS | Nausea, cramps, steatorrhea. ↑ risk of cholelithiasis due to CCK inhibition.                     |

**Sulfasalazine**

|                 |                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| MECHANISM       | A combination of sulfapyridine (antibacterial) and 5-aminosalicylic acid (anti-inflammatory). Activated by colonic bacteria. |
| CLINICAL USE    | Ulcerative colitis, Crohn disease (colitis component).                                                                       |
| ADVERSE EFFECTS | Malaise, nausea, sulfonamide toxicity, reversible oligospermia.                                                              |

**Loperamide**

|                 |                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------|
| MECHANISM       | Agonist at μ-opioid receptors; slows gut motility. Poor CNS penetration (low addictive potential). |
| CLINICAL USE    | Diarrhea.                                                                                          |
| ADVERSE EFFECTS | Constipation, nausea.                                                                              |

**Ondansetron**

|                 |                                                                                        |
|-----------------|----------------------------------------------------------------------------------------|
| MECHANISM       | 5-HT <sub>3</sub> antagonist; ↓ vagal stimulation. Powerful central-acting antiemetic. |
| CLINICAL USE    | Control vomiting postoperatively and in patients undergoing cancer chemotherapy.       |
| ADVERSE EFFECTS | Headache, constipation, QT interval prolongation, serotonin syndrome.                  |

**Metoclopramide**

|                 |                                                                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MECHANISM       | D <sub>2</sub> receptor antagonist. ↑ resting tone, contractility, LES tone, motility, promotes gastric emptying. Does not influence colon transport time.                                                                                                                        |
| CLINICAL USE    | Diabetic and postsurgery gastroparesis, antiemetic, persistent GERD..                                                                                                                                                                                                             |
| ADVERSE EFFECTS | ↑ parkinsonian effects, tardive dyskinesia. Restlessness, drowsiness, fatigue, depression, diarrhea. Drug interaction with digoxin and diabetic agents. Contraindicated in patients with small bowel obstruction or Parkinson disease (due to D <sub>2</sub> -receptor blockade). |

**Orlistat**

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| MECHANISM       | Inhibits gastric and pancreatic lipase → ↓ breakdown and absorption of dietary fats. |
| CLINICAL USE    | Weight loss.                                                                         |
| ADVERSE EFFECTS | Steatorrhea, ↓ absorption of fat-soluble vitamins.                                   |

**Laxatives**

Indicated for constipation or patients on opiates requiring a bowel regimen

**Bulk-forming laxatives**

Psyllium, methylcellulose

|                 |                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------|
| MECHANISM       | Soluble fibers; draw water into gut lumen, forming a viscous liquid that promotes peristalsis |
| ADVERSE EFFECTS | Bloating                                                                                      |

**Osmotic laxatives**

Magnesium hydroxide, magnesium citrate, polyethylene glycol, lactulose

|                 |                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MECHANISM       | Provide osmotic load to draw water into GI lumen<br>Lactulose also treats hepatic encephalopathy because gut flora degrade it into metabolites (lactic acid, acetic acid) that promote nitrogen excretion as NH <sub>4</sub> <sup>+</sup> |
| ADVERSE EFFECTS | Diarrhea, dehydration; may be abused by bulimics                                                                                                                                                                                          |

**Stimulants**

Senna

|                 |                                                 |
|-----------------|-------------------------------------------------|
| MECHANISM       | Enteric nerve stimulation → colonic contraction |
| ADVERSE EFFECTS | Diarrhea, melanosis coli                        |

**Emollients**

Docusate

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| MECHANISM       | Osmotic draw into lumen → ↑ water absorption by stool |
| ADVERSE EFFECTS | Diarrhea                                              |

**Aprepitant**

|              |                                                                    |
|--------------|--------------------------------------------------------------------|
| MECHANISM    | Substance P antagonist. Blocks NK <sub>1</sub> receptors in brain. |
| CLINICAL USE | Antiemetic for chemotherapy-induced nausea and vomiting.           |

▶ NOTES